

Dominant Singularities, Sharp Euler Extrema, and Lambert Polynomial Zeros

A rigorous continuation of the ProveIt
polylogarithm and harmonic-order programs

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Source baseline. [Polylogarithms and their Arithmetic Bridges](#), its harmonic-order continuation, and the six archives present in `docs/incoming` at the pinned revision.

Research status. Theorems have ordinary mathematical proofs. Finite algebraic certificates and high-precision numerical diagnostics have separate roles. The statements are not presented as Lean formalizations or as claims of priority over the entire literature.

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Abstract

This article continues two analytic programs in the ProveIt polylogarithm manuscript: sharp Euler remainder estimates at arbitrary real orders, and the motion of real zeros under harmonic and Lerch deformation. For the optimal normalized remainder C_N , we prove eventual exact reduction to the outer-order boundary $a = 0$, uniqueness of the optimizing inner order, and the conjectured law $C_N = 1 + cN^{-p} + O(N^{-p-\delta})$, with explicit positive constants. For the pointwise remainder kernel, we prove global convergence of all maximizers to a unique interior scaled limit, a convergent expansion in $1/N$, and eventual strict decrease of its optimum. The first later truncation, $N = 2$, has an exact algebraic optimum of degree seven. For the inverse equation $e^U = A + tU$, an exact rectangular barrier selects the relevant analytic sheet and its two dominant square-root singularities for every $A > 1$. This makes the repository's conditional velocity asymptotic unconditional, yields all-order oscillatory expansions, bounded sign gaps, convergent boundary identities, and an explicit limiting distribution of the zero-velocity lattice parameters. The inverse series is identified with earlier work of Kalugin and Jeffrey, and the coefficient polynomials with Cohen's Lambert polynomials. For the latter we prove real-rootedness, strict interlacing, an all-orders fixed-parameter expansion of the largest roots with the polynomial coefficient degrees conjectured by Cohen, and a Bessel limit at the smallest-root endpoint. The first two largest-root corrections are explicit combinations of zeta values. All claims are separated from numerical evidence and from questions that remain open. Exact algebraic verification, numerical diagnostics, source provenance, and integration notes accompany the article.

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1 Research scope, results, and relation to the baseline

The organizing question is how a global extremum or a global inverse sheet can be recovered from a local asymptotic calculation. In both problems considered here, the local calculation was substantially available in the source material, but it did not by itself settle the global conclusion. For Euler remainders, a trial family on the axis gave the right candidate correction to the optimal constant; one still had to exclude all other order parameters. For harmonic zero velocities, formal critical points and local square-root expansions were known; one still had to prove that the selected inverse branch actually reaches exactly the proposed dominant pair. The proofs below supply those global steps.

1.1 A fixed and reproducible source comparison

The repository baseline is revision `4c173c06cc32c9cea554b39be1837ad2ae897fc1`, dated October 10, 2026. The review covered the modular manuscript, its editorial ledger and validation notes, the harmonic-order geometry continuation, and all six archives then present in `docs/incoming`. The provenance manifest records the paths and Git blob hashes of the retrieved sources. Statements about what was already proved or left open refer to this revision, rather than to an evolving branch [1, 2, 3, 4].

Several substantial problems have already been settled in that baseline. In particular, the sharp constant uniform in *all* positive real orders and *all* truncation indices is already known; so are the S_2 and S_4 reductions and the certified radial double-extremum phenomenon. They are not reported here as fresh results. The unresolved S_6 reduction is not claimed. The selected targets are the remaining global large-index Euler problem and the dominant-sheet question for diagonal harmonic velocities, together with consequences naturally forced by their proofs.

1.2 The main analytic conclusions

Use the precise normalization of section 2. The first result is

$$C_N = \max_{b>0} R_N(0, b) \quad \text{for all sufficiently large } N,$$

with a unique maximizing order b_N . Put

$$d = \log(3/2), \quad p = \frac{\log 2}{d}, \quad q = \frac{\log 3}{\log 2}, \quad c = \frac{p^p}{(p+1)^{p+1}}, \quad \delta = \frac{\log(10/9)}{d}.$$

Then

$$b_N = \frac{\log(Nq)}{d} + O(N^{-\delta}), \tag{1.1}$$

$$C_N = 1 + cN^{-p} + O(N^{-p-\delta}). \tag{1.2}$$

Here $p = 1.70951129135\dots$ and $c = 0.167947796543\dots$. The positive-quadrant supremum is approached as $a \downarrow 0$ and is not attained there. The proof also gives a necessary scale for any near-optimal outer order, sharpened to a necessary and sufficient criterion and a joint profile with exact linear loss $\sqrt{\pi}$. It uses probability kernels and positive Taylor remainders, so no cancellation-sensitive finite Euler sum is needed as a proof premise.

The pointwise kernel optimum M_N is a different quantity. Its limit is

$$M_\infty = 1.053861930362711626\dots > 1,$$

whereas $C_N \rightarrow 1$. Every global kernel maximizer satisfies $Nr_N \rightarrow 3.941064831647250668\dots$ and $y_N \rightarrow 0.145303859259358424\dots$. The optimized kernel value and location have convergent expansions in $1/N$ near zero. The exact degree-seven answer for M_2 gives a finite-index anchor for this asymptotic theory.

For the harmonic problem, let $A = 1 + h > 1$ encode the lattice spacing. There is a unique $\theta \in (0, \pi)$ such that

$$\log A = 1 - \theta \cot \theta + \log \frac{\theta}{\sin \theta}. \quad (1.3)$$

Writing $a_* = 1 - \theta \cot \theta$, $u_* = a_* + i\theta$, and $R = e^{a_*}$, the selected inverse of $e^U = A + tU$ has precisely the two singularities $Re^{\pm i\theta}$ on its convergence circle. If $v_n(A)$ is the initial velocity of the n th diagonal zero, then

$$v_n(A) = \frac{R^{-n}}{\sqrt{\pi} n^{3/2}} \Re \left(e^{-in\theta} \sqrt{-2u_*} \right) + O_A(R^{-n} n^{-5/2}),$$

with an explicitly selected square root. The article gives its next coefficient and an algorithm for every subsequent one. The error is additive; no relative approximation is asserted near a cosine zero.

The same phase describes the distribution of the positive zeros of the velocity polynomials. In the angle coordinate their limiting distribution is uniform on $(0, \pi)$. In the logarithmic coordinate $L = \log A$ its density is

$$\frac{\theta}{\pi((1 - \theta \cot \theta)^2 + \theta^2)}.$$

Thus complex singularity geometry determines both oscillation in the index and the distribution of real parameters where an initial velocity vanishes.

The Lambert-polynomial identification also leads to an all-orders solution of the fixed-parameter extreme-root questions in Cohen's Conjecture 7.1 [6]. If $X_{n,k,j}$ is the j th largest positive root of $L_{n,k}$, then for fixed positive integers j, k and any fixed M ,

$$X_{n,k,j} = \frac{n+k-1}{j} + \log n + \gamma - H_{j-1} + \sum_{m=1}^M \frac{C_{j,m}(k-1)}{n^m} + O_{j,k,M}(n^{-M-1}). \quad (1.4)$$

Each $C_{j,m}$ is a polynomial of degree exactly $m-1$. The first largest-root coefficient is

$$C_{1,1} = -\frac{3}{2} - \zeta(2) - \zeta(3),$$

and the next coefficient is given explicitly in section 14. Euler's constant occurs only in the leading shift; every later coefficient lies in a specified algebra generated by ordinary zeta values. A different endpoint scaling gives $n^2 x_{n,k,j} \rightarrow z_{k,j}^2/2$ for the j th smallest positive root, where $z_{k,j}$ is a positive zero of J_k . The bulk, large-root, and Bessel limits have separately stated regimes.

1.3 Attribution and the meaning of a result here

Kalugin and Jeffrey's 2012 Lambert- W series already contains the inverse coefficient array and the convergence-radius criterion after an explicit change of variables [5]. Cohen's Lambert polynomials are the same coefficient polynomials up to a sign [6]. These are substantive identifications, not merely analogies. They correct a gap in the source comparison and make it possible to connect the repository's questions to the existing literature. The barrier proof, its detailed continuation statement, and the derived velocity and zero-distribution results are presented with this antecedent made explicit.

Throughout the article, “proved” means proved in ordinary mathematics with the hypotheses stated. The exact polynomial computations have independent finite verifiers. Numerical calculations illustrate scales, locate examples, and test formulas; they are not substituted for global analytic arguments. No assertion of literature-wide priority or machine-checked formalization is made. The final sections give precise proposed source amendments and distinguish completed targets from the remaining finite-index, edge-scaling, and arithmetic questions.

2 A common probability representation for Euler remainders

For $a \geq 0$ and $b > 0$, start in $|z| < 1$ with

$$F_{a,b}(z) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad H_0^{(b)} = 0. \quad (2.1)$$

At integer orders this is a depth-two polylogarithmic family; the probability representation below also handles arbitrary real orders. Let T_a, U_b be independent gamma variables of rate one and shapes a, b , with $T_0 = 0$. Put $X = e^{-T_a}$ and $V = e^{-U_b}$. Their moments are

$$\mathbb{E} X^{m-1} = m^{-a}, \quad \mathbb{E} V^{m-1} = m^{-b} \quad (m \geq 1).$$

In particular,

$$\frac{H_{n-1}^{(b)}}{n^a} = \mathbb{E} \left[X^{n-1} \frac{1 - V^{n-1}}{1 - V} \right]. \quad (2.2)$$

The quotient at $n = 1$ means zero, and otherwise is a finite geometric sum. Summing in z gives the analytic continuation

$$F_{a,b}(z) = \mathbb{E} \frac{z^2 X}{(1 - zX)(1 - zXV)}, \quad z \in \mathbb{C} \setminus [1, \infty). \quad (2.3)$$

Indeed the integrand is uniformly bounded on compact subsets of the slit plane, and the identity holds first in the unit disk. Thus the Gaussian value $g_{a,b} = \Im F_{a,b}(i)$ is unambiguous even when the corresponding unregularized boundary series fails to converge. Explicitly,

$$g_{a,b} = -\mathbb{E} \frac{X^2(1 + V)}{(1 + X^2)(1 + X^2V^2)}. \quad (2.4)$$

For an integer $N \geq 1$, define

$$f_n = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \quad d_j = \sum_{n=0}^j (-1)^n \binom{j}{n} f_n, \quad (2.5)$$

$$E_N = \sum_{j=0}^{N-1} \frac{d_j}{2^{j+1}}, \quad R_N(a, b) = 2^N (E_N - g_{a,b}), \quad C_N = \sup_{a,b>0} R_N(a, b). \quad (2.6)$$

The sign convention is essential: E_N is an upper approximation to the negative value $g_{a,b}$. Euler’s classical series transformation is the background for this normalization [10]; here all identities can be obtained directly from finite sums.

Proposition 2.1 (Exact positive remainder and a boundary extension). *With*

$$W_N(r) = \frac{(1-r)^N}{1+r}, \quad K_N(r, v) = \frac{W_N(rv^2) - W_N(r)}{1-v},$$

one has, for every $a \geq 0$, $b > 0$, and $N \geq 1$,

$$R_N(a, b) = \mathbb{E} K_N(X^2, V) = \mathbb{E} \frac{W_N(e^{-2(T_a+U_b)}) - W_N(e^{-2T_a})}{1 - e^{-U_b}} > 0. \quad (2.7)$$

Moreover $d_0 = 0$, $d_j < 0$ for $j \geq 1$, and E_N decreases to $g_{a,b}$. The remainder is continuous as $a \downarrow 0$ at every fixed $b > 0$.

Proof. The finite binomial identity from (2.2) gives

$$d_j = \mathbb{E} \frac{(1 - X^2)^j - (1 - X^2 V^2)^j}{1 - V}.$$

Its signs are immediate. Summing the finite geometric series defining E_N and using (2.4) proves (2.7). Decrease of W_N gives strict positivity. The contraction estimate already established in the manuscript is

$$0 \leq K_N(r, v) \leq \frac{1+v}{1+v^2} \leq \frac{1+\sqrt{2}}{2}. \quad (2.8)$$

For a self-contained verification when $N \geq 2$, use the power envelope in lemma 6.1 and the positive geometric expansion of W_N in lemma 6.2; its $m = 2$ envelope is precisely the first upper bound. For $N = 1$, subtraction from that upper bound leaves

$$\frac{(1+v)(1-r)(1-rv^2)}{(1+v^2)(1+rv^2)(1+r)} \geq 0.$$

Thus $E_N - g_{a,b} = O(2^{-N})$, which proves convergence. Gamma variables T_a tend in probability to zero as $a \downarrow 0$; bounded convergence using (2.8) proves the boundary continuity. $\square$

The representation and contraction bound are inherited results [1, 4]. The following short conditioning argument supplies a useful direct proof of another inherited bound, without the signed-measure machinery needed elsewhere in the manuscript.

Lemma 2.2 (The outer-order region $a \geq 1$). *For all $a \geq 1$, $b > 0$, and $N \geq 1$, one has $R_N(a, b) < 1$.*

Proof. Decompose $T_a = T_1 + T_{a-1}$ independently. Since e^{-T_1} is uniform on $(0, 1)$, condition on $Z = e^{-T_{a-1}} \in (0, 1]$ and $V = v \in (0, 1)$. The conditional expectation of the kernel is

$$\begin{aligned} & \int_0^1 K_N(Z^2 u^2, v) du \\ &= \frac{v^{-1} \int_0^{Zv} W_N(t^2) dt - \int_0^Z W_N(t^2) dt}{Z(1-v)} \\ &\leq \frac{1}{Zv} \int_0^{Zv} W_N(t^2) dt < 1. \end{aligned}$$

The first inequality uses $W_N \geq 0$, and the second uses $W_N(t^2) < 1$ for $t > 0$. Taking expectations proves the claim. $\square$

2.1 Three constants that must be distinguished

Write

$$C_N^{\text{ax}} = \max_{b>0} R_N(0, b), \quad M_N = \max_{(r,v) \in [0,1]^2} K_N(r, v),$$

where existence of the axis maximum is proved below. The boundary continuity and the probability representation give

$$C_N^{\text{ax}} \leq C_N \leq M_N. \quad (2.9)$$

The already known all-index constant is $C_* := \sup_{N \geq 1} C_N = 1.13656\dots$, attained only after adjoining $a = 0$ and then at $N = 1$ and a unique inner order [1, 4]. The present global theorem proves $C_N^{\text{ax}} = C_N$ for sufficiently large N and $C_N \rightarrow 1$. The separate kernel theorem proves $M_N \rightarrow M_\infty > 1$. Maximization before and after gamma averaging cannot be interchanged without proof.

3 Global optimization of the real-order Euler remainder

The matching upper asymptotic was explicitly left open as Conjecture `conj:euler-rate` in the incoming uniform-bounds report [3]. Its axis lower bound already identified the candidate exponent and leading amplitude. The additional ingredients here are global localization over both orders, a positive loss that forces the outer order to the boundary, and uniform derivative control for the inner-order optimization. The region $a \geq 1$ is excluded by lemma 2.2, whose direct proof was given above.

3.1 Statement

Put

$$d = \log(3/2), \quad p = \frac{\log 2}{d}, \quad q = \frac{\log 3}{\log 2}, \quad c = (1 - q^{-1})q^{-p}, \quad \delta = \frac{\log(10/9)}{d} > 0,$$

and

$$\hat{b}_N = \frac{\log(Nq)}{d}.$$

Thus $p = 1.7095112913514547\dots$, $c = 0.1679477965425128\dots$, and $\delta = 0.2598510045646633\dots$. For independent gamma variables of rate one, with shape zero interpreted as the point mass at zero, define

$$W_N(z) = \frac{(1-z)^N}{1+z}, \quad F_N(t) = W_N(e^{-2t}), \quad J_N(a) = \mathbb{E} F_N(T_a).$$

The existing all-order Euler formula is

$$R_N(a, b) = \mathbb{E} \frac{F_N(T_a + U_b) - F_N(T_a)}{1 - e^{-U_b}}, \quad a \geq 0, \quad b > 0. \quad (3.1)$$

In particular $J_N(0) = F_N(0) = 0$.

Theorem 3.1 (Eventual exact axis reduction). *There is an integer N_0 such that for every $N \geq N_0$,*

$$C_N := \sup_{a,b>0} R_N(a, b) = \max_{b>0} R_N(0, b). \quad (3.2)$$

The maximum on the right is attained at exactly one b_N , and

$$b_N = \widehat{b}_N + O(N^{-\delta}), \quad (3.3)$$

$$C_N = 1 + cN^{-p} + O(N^{-p-\delta}). \quad (3.4)$$

For $N \geq N_0$ the supremum on the positive quadrant is not attained; on the domain $a \geq 0, b > 0$ its unique maximizer is $(0, b_N)$. Moreover $C_{N+1} < C_N$ for every sufficiently large N .

Theorem 3.2 (Scale of near maximizers). *If $a_N, b'_N > 0$ and*

$$C_N - R_N(a_N, b'_N) = o(N^{-p}),$$

then

$$b'_N - \widehat{b}_N \longrightarrow 0, \quad a_N = o\left(N^{-(p-1/2)} \log N\right).$$

The same assertion holds with $a_N = 0$ allowed. This accuracy requirement is essential: the incoming moving-order phase law already gives many sequences with $R_N \rightarrow 1$ and a_N increasing to infinity.

3.2 Elementary bounds and a gamma expansion

The function F_N increases from zero to one. Direct differentiation gives $0 \leq F'_N \leq 4$, uniformly in N , and F'_N tends uniformly to zero on every fixed bounded interval. These facts also follow from

$$F'_N(t) = \frac{2z(1-z)^{N-1}\{N+1+(N-1)z\}}{(1+z)^2}, \quad z = e^{-2t}.$$

For $N \geq 2$ use $Nz(1-z)^{N-1} \leq 1$; for $N = 1$ the bound is immediate.

For $b > 1$ expand $(1 - e^{-U_b})^{-1}$ positively. Exponential tilting changes U_b to a gamma variable $U_b^{(m)}$ of rate m , with total weight m^{-b} . Consequently

$$R_N(a, b) = \sum_{m \geq 1} m^{-b} \{\mathbb{E} F_N(T_a + U_b^{(m)}) - J_N(a)\}. \quad (3.5)$$

Every term in braces lies in $[0, 1]$, and the term $m = 1$ is $J_N(a+b) - J_N(a)$. Therefore

$$R_N(a, b) \leq \zeta(b), \quad R_N(a, b) \leq J_N(a+b) - J_N(a) + \zeta(b) - 1. \quad (3.6)$$

For $b \geq 2$ the elementary integral estimate gives

$$\zeta(b) - 1 \leq 2^{-b} \left(1 + \frac{2}{b-1}\right) \leq 3 \cdot 2^{-b}. \quad (3.7)$$

We will use the sharper factor 2 once $b \rightarrow \infty$.

The function W_N is convex and

$$W_N(z) = 1 - (N+1)z + e_N(z), \quad 0 \leq e_N(z) \leq B_N z^2, \quad B_N = \frac{N^2 + N + 2}{2}. \quad (3.8)$$

Furthermore e_N is nondecreasing on $[0, 1]$. For completeness, $W'_N(0) = -(N+1)$ and $0 \leq W''_N(z) \leq W''_N(0) = N^2 + N + 2$. For $N \geq 2$ the latter follows by differentiating $(1-z)^N/(1+z)$ twice and bounding each positive term at zero; for $N = 1$ use $H_1(z) = 2/(1+z) - 1$. Convexity gives $W'_N(z) \geq W'_N(0)$, proving $e'_N \geq 0$.

Write $S_j(b) = \sum_{m \geq j+1} m^{-b}$. Substitution of (3.8) in (3.1), with $x = e^{-T_a}$ and $v = e^{-U_b}$, gives the exact identity

$$R_N(a, b) = \zeta(b)(1 - J_N(a)) - (N + 1)3^{-a}S_2(b) + \eta_N(a, b), \quad (3.9)$$

where

$$\eta_N(a, b) = \mathbb{E} \frac{e_N(x^2 v^2)}{1 - v}, \quad 0 \leq \eta_N(a, b) \leq B_N 5^{-a} S_4(b). \quad (3.10)$$

This is valid for $b > 1$. Gamma addition couples $T_{a_2} \geq T_{a_1}$ when $a_2 > a_1$. Since e_N is nondecreasing, it also proves the important exact monotonicity

$$a \mapsto \eta_N(a, b) \quad \text{is nonincreasing.} \quad (3.11)$$

3.3 Uniform localization of every competitive pair

Call a pair competitive if

$$R_N(a, b) \geq 1 + \frac{c}{2} N^{-p}. \quad (3.12)$$

The axis lower asymptotic already proved in the incoming report implies that competitive pairs exist for all sufficiently large N , including positive a by continuity at $a = 0$. By lemma 2.2, every competitive pair has $0 \leq a < 1$. Set $\ell = \log N$ and $L = \ell/2$.

Lemma 3.3 (A fixed logarithmic band). *There are fixed constants $M_-, M_+ > 0$ such that for all sufficiently large N , every competitive pair satisfies*

$$0 \leq a < 1, \quad \widehat{b}_N - M_- \leq b \leq \widehat{b}_N + M_+.$$

One may take $M_- = 10$ and $M_+ = 5$ after increasing the unspecified threshold in N .

Proof. We first prove that $R_N(a, b) < 1$ whenever $0 \leq a \leq 1$ and $b \leq \widehat{b}_N - 10$, for all sufficiently large N .

Bounded inner orders, $0 < b \leq 2$. Gamma addition permits a coupling with $T_a \leq T_1$ and $U_b \leq U_2$, independently. The mean-value formula and $u/(1 - e^{-u}) \leq 1 + u$ imply

$$\frac{F_N(T_a + U_b) - F_N(T_a)}{1 - e^{-U_b}} \leq (1 + U_2) \sup_{0 \leq t \leq T_1 + U_2} F'_N(t).$$

The right side is independent of the smaller shapes, bounded by the integrable variable $4(1 + U_2)$, and tends to zero almost surely. Dominated convergence proves uniform convergence $R_N \rightarrow 0$ on this rectangle.

Small growing orders, $2 \leq b \leq \ell/4$. The second inequality of (3.6), stochastic monotonicity of J_N , and $\zeta(2) - 1 \leq 3/4$ give

$$R_N(a, b) \leq J_N(1 + \ell/4) + 3/4.$$

The first term tends to zero. Indeed a gamma variable of shape $1 + \ell/4$ divided by ℓ tends to $1/4$ in probability, by its mean and variance; on $t \leq 3\ell/8$, $F_N(t) \leq \exp(-N^{1/4})$, and the probability of the complement tends to zero. Hence this whole range eventually has $R_N < 1$.

Intermediate growing orders, $\ell/4 \leq b \leq 2\ell$. Put $s = a + b = \alpha\ell$. For large N , $1/4 \leq \alpha \leq 2 + 1/\ell$. For $t \in [L, L + 1]$,

$$1 - F_N(t) \geq d_0 := 1 - \exp(-e^{-2}) > 0.$$

Since $s \geq 1$, the gamma density on that interval gives

$$1 - J_N(s) \geq d_0 e^{-1} N^{-1/2} \frac{L^{s-1}}{\Gamma(s)}. \quad (3.13)$$

The real Stirling bound $\Gamma(s) \leq A s^{s-1/2} e^{-s}$ for $s \geq 1$, with any fixed valid constant A , yields uniformly

$$1 - J_N(s) \geq A_1 \ell^{-1/2} \exp\{-\ell I(\alpha)\}, \quad I(\alpha) = \frac{1}{2} - \alpha + \alpha \log(2\alpha).$$

Comparison with $2^{-b} = 2^a \exp(-\alpha \ell \log 2)$ shows

$$\frac{1 - J_N(a + b)}{2^{-b}} \geq A_2 \ell^{-1/2} \exp\{\ell Q(\alpha)\}, \quad Q(\alpha) = \alpha - \frac{1}{2} - \alpha \log \alpha.$$

This tends uniformly to infinity. To check uniformity explicitly, Q is concave and positive at both endpoints of $[1/4, 2]$:

$$Q(1/4) = (\log 4 - 1)/4 > 0, \quad Q(2) = 3/2 - 2 \log 2 > 0.$$

By continuity and concavity, its minimum on $[1/4, 2 + 1/\ell]$ is bounded below by a fixed positive number for all large ℓ . For example $Q \geq 1/40$ on $[1/4, 21/10]$; the last upper endpoint contains the required interval when $\ell \geq 10$. Thus (3.7) gives $1 - J_N(a + b) > \zeta(b) - 1$ uniformly in this range. Equation (3.6) now proves $R_N(a, b) < 1$.

Upper logarithmic range, $2\ell \leq b \leq \hat{b}_N - 10$. Average (3.8) with a gamma shape $s = a + b$ to obtain

$$1 - J_N(s) \geq (N + 1)3^{-s} - B_N 5^{-s}.$$

For $s \geq 2\ell$, the ratio of the second term to the first is at most

$$\frac{B_N}{N + 1} (3/5)^{2\ell} = O\left(N^{1-2\log(5/3)}\right) = o(1),$$

since $2\log(5/3) > 1$. With $0 \leq a \leq 1$ this gives, for large N ,

$$1 - J_N(a + b) \geq \frac{N}{4} 3^{-b}.$$

On the other hand $\zeta(b) - 1 \leq 2 \cdot 2^{-b}$ here. At $b \leq \hat{b}_N - 10$,

$$\frac{(N/4)3^{-b}}{2^{-b}} \geq \frac{(3/2)^{10}}{4q} > 2.$$

Thus the gamma loss again exceeds $\zeta(b) - 1$, and (3.6) proves $R_N < 1$.

These four ranges cover every $b \leq \hat{b}_N - 10$ for all large N , because $1/d > 2$. They prove the lower edge of the band. For a competitive pair, the first inequality of (3.6) and $\zeta(b) - 1 \leq 2 \cdot 2^{-b}$ imply

$$\frac{c}{2} N^{-p} \leq 2 \cdot 2^{-b}.$$

Using $2^{-\hat{b}_N} = q^{-p} N^{-p}$, this gives

$$b - \hat{b}_N \leq \log_2 \frac{4}{1 - q^{-1}} < 5.$$

This proves the upper edge and completes the proof. $\square$

3.4 A positive loss that forces the boundary maximum

Lemma 3.4 (Uniform lower bound for the outer-order loss). *There is a fixed $D > 0$ such that, for all sufficiently large N ,*

$$J_N(a) \geq Da \frac{N^{-1/2}}{\log N} \quad (0 \leq a \leq 1).$$

Proof. The assertion is immediate at $a = 0$. For $0 < a \leq 1$, integrate the positive gamma representation over $t \in [L+1, L+2]$. On this interval, Bernoulli's inequality gives

$$F_N(t) \geq F_N(L+1) \geq h_0 := \frac{1 - e^{-2}}{2} > 0.$$

Log-convexity of Γ and $\Gamma(1) = \Gamma(2) = 1$ imply $\Gamma(a+1) \leq 1$, so $1/\Gamma(a) = a/\Gamma(a+1) \geq a$. For large N , also $t^{a-1} \geq 1/t \geq 1/(L+2)$ on the interval, and $e^{-t} \geq e^{-2}N^{-1/2}$. Therefore

$$J_N(a) \geq h_0 e^{-2} \frac{aN^{-1/2}}{L+2},$$

which proves the claimed uniform bound. $\square$

Proposition 3.5 (Strict comparison with the axis). *For all sufficiently large N and every $b \in [\hat{b}_N - 10, \hat{b}_N + 5]$,*

$$R_N(0, b) - R_N(a, b) \geq d_{\text{loss}} a \frac{N^{-1/2}}{\log N} \quad (0 \leq a \leq 1), \quad (3.14)$$

for a fixed constant $d_{\text{loss}} > 0$.

Proof. Subtract (3.9) at a and at zero, and use (3.11), Lemma 3.4, $\zeta(b) \geq 1$, and $1 - 3^{-a} \leq a \log 3$. This gives

$$R_N(a, b) - R_N(0, b) \leq a \left\{ -D \frac{N^{-1/2}}{\log N} + (N+1)(\log 3) S_2(b) \right\}.$$

Uniformly in the stated band, $S_2(b) = O(3^{-b})$ and $N3^{-b} = O(N^{-p})$. Since $p > 1/2$, the positive term is smaller than half the negative one for all sufficiently large N . Take $d_{\text{loss}} = D/2$. $\square$

3.5 Axis optimization with controlled derivatives

On the axis, the exact kernel is

$$R_N(0, b) = \mathbb{E} k_N(e^{-U_b}), \quad k_N(v) = \frac{(1+v)(1-v^2)^{N-1}}{1+v^2}.$$

For every fixed $N \geq 2$ this is continuous in $b > 0$, tends to zero as $b \downarrow 0$, and tends to one as $b \rightarrow \infty$. The last limit follows from bounded convergence under gamma addition, or directly from the moment expansion below. That expansion also shows $R_N(0, b) > 1$ for sufficiently large b , with N fixed. Thus its global maximum is attained at a finite positive b .

Proposition 3.6 (Unique nondegenerate axis maximum at every index). *For every integer $N \geq 1$, $b \mapsto R_N(0, b)$ has exactly one stationary point. It is a nondegenerate global maximum, of height strictly greater than one.*

Proof. For $N \geq 2$, the sign of $k'_N(v)$ is that of

$$h_N(v) = 1 - (2N + 1)v + v^2 + (3 - 2N)v^3.$$

Its derivative is strictly negative on $[0, 1]$, while $h_N(0) = 1$ and $h_N(1) = 4 - 4N < 0$. Thus $\tilde{k}_N(t) = k_N(e^{-t})$ increases from zero to a unique maximum greater than one, then decreases to one. For $N = 1$ the same shape statement holds with value one at both endpoints; use $k'_1(v) = (1 - 2v - v^2)/(1 + v^2)^2$.

Fix b and write $M = \mathbb{E} \tilde{k}_N(U_b) = R_N(0, b)$. Differentiation under the gamma integral gives

$$R'_N(0, b) = \mathbb{E}\{(\tilde{k}_N(U_b) - M) \log U_b\},$$

where the notation here means the derivative in b of the axis function. If $M \leq 1$, then $\tilde{k}_N(t) - M$ has exactly one finite sign change, from negative to positive, at a point t_0 . Subtracting $\log t_0$ in this expectation makes its integrand strictly positive except at the crossing, so the derivative is positive. For $N = 1$, $M > 1$ automatically.

At any stationary point we therefore have $M > 1$. The function $\tilde{k}_N(t) - M$ then has exactly two crossings $t_1 < t_2$, with signs negative, positive, negative. Both its gamma average and its logarithmic first moment vanish at the stationary point. A second differentiation consequently gives

$$\begin{aligned} \frac{d^2}{db^2} R_N(0, b) &= \mathbb{E}\{(\tilde{k}_N(U_b) - M)(\log U_b)^2\} \\ &= \mathbb{E}\{(\tilde{k}_N(U_b) - M)(\log U_b - \log t_1)(\log U_b - \log t_2)\} < 0. \end{aligned}$$

The last integrand is strictly negative away from the crossings. All differentiations are justified by boundedness of $\tilde{k}_N$ and the finite logarithmic moments of gamma laws locally in $b > 0$.

The endpoint limits and a value above one guarantee at least one interior maximum. Every stationary point has negative second derivative, so there cannot be two: between two strict maxima there would be a stationary minimum (or the derivative would have to cross from negative to positive). This proves the assertion. $\square$

The incoming Taylor identity, restated to track its derivatives, is

$$R_N(0, b) = 1 + 2^{-b} - N3^{-b} - N4^{-b} + \varepsilon_N(b), \quad (3.15)$$

where

$$0 \leq \varepsilon_N(b) \leq A_N(5^{-b} + 6^{-b}), \quad A_N = (N^2 - N + 2)/2. \quad (3.16)$$

Indeed, for $G_N(z) = (1 - z)^{N-1}/(1 + z)$, $G_N(z) = 1 - Nz + r_N(z)$ with $0 \leq r_N(z) \leq A_N z^2$; multiply by $1 + v$, set $z = v^2$, and integrate the gamma moments.

For $k = 0, 1, 2$, the remainder has the derivative bound

$$|\varepsilon_N^{(k)}(b)| \leq A_k N^2 5^{-b} \quad (b \geq 1), \quad (3.17)$$

with constants A_k independent of N, b . To see this without differentiating an inequality, write the exact nonnegative remainder as

$$\varepsilon_N(b) = \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} e^{-t} w_N(t) dt, \quad 0 \leq w_N(t) \leq A_N(e^{-4t} + e^{-5t}).$$

The first two derivatives of the gamma density multiply it by

$$\log t - \psi(b), \quad (\log t - \psi(b))^2 - \psi_1(b).$$

After exponential tilting to rate $m \in \{5, 6\}$, the score has mean $-\log m$ and variance $\psi_1(b) \leq \psi_1(1) = \pi^2/6$. Cauchy–Schwarz therefore bounds the absolute first score moment by $\log m + \pi/\sqrt{6}$, and the absolute second derivative factor by $(\log m)^2 + \pi^2/3$. These bounds prove (3.17). Differentiation is justified by the same locally integrable domination.

For bounded real s , equations (3.15)–(3.17) give, uniformly with two derivatives in s ,

$$N^p \{R_N(0, \hat{b}_N + s) - 1\} = f(s) + O(N^{-\delta}), \quad (3.18)$$

where

$$f(s) = q^{-p} 2^{-s} - q^{-(p+1)} 3^{-s}.$$

For the error exponents, $N 4^{-\hat{b}_N} = O(N^{-p-(p-1)})$ and $N^2 5^{-\hat{b}_N} = O(N^{-p-\delta})$, with $p-1 > \delta > 0$. The same statements hold after up to two derivatives.

The function f has its unique global maximum at $s = 0$:

$$f(0) = c, \quad f'(0) = 0, \quad f''(0) = -(\log 2) d q^{-p} < 0.$$

Its derivative has exactly one zero because its two exponential terms have distinct rates.

3.6 Completion of the proofs

Proof of Theorem 3.1. The axis value at $\hat{b}_N$ is $1 + cN^{-p} + O(N^{-p-\delta})$, so its maximum exceeds $1 + (c/2)N^{-p}$ for all sufficiently large N . Every pair that could exceed this axis maximum is therefore competitive. Lemma 3.3 places it in $0 \leq a < 1$ and the fixed logarithmic band. Proposition 3.5 gives $R_N(a, b) \leq R_N(0, b)$ there, strictly for $a > 0$. Pairs outside that competitive class cannot be maximizers. This proves the exact global reduction (3.2); continuity at $a = 0$ shows that the same value is the supremum over $a, b > 0$.

Every axis maximizer also lies in the band of Lemma 3.3. Uniform convergence in (3.18), compactness of the band, and the unique maximum of f force every maximizing offset $b - \hat{b}_N$ to tend to zero. In a fixed small neighborhood of zero, f'' is bounded above by a strictly negative constant. The uniform second derivative version of (3.18) makes the axis function strictly concave there for all large N . Since all of its global maxima lie there, its global maximum is unique. Denote it by b_N . Its first derivative vanishes, so $f'(b_N - \hat{b}_N) = O(N^{-\delta})$. The nonzero value of $f''(0)$ and the mean-value theorem imply (3.3). Equation (3.18) then gives (3.4).

The strict axis comparison proves nonattainment in the positive quadrant and uniqueness on the closed axis domain. Finally, for each finite $b > 0$, $k_{N+1}(v) = (1 - v^2)k_N(v) < k_N(v)$ on $(0, 1)$. At the finite maximizer of the $(N+1)$ st axis problem,

$$C_{N+1} = R_{N+1}(0, b_{N+1}) < R_N(0, b_{N+1}) \leq C_N.$$

This applies whenever both adjacent indices have the axis reduction. $\square$

Proof of Theorem 3.2. The assumed accuracy makes the pair competitive for all large N . It therefore lies in the fixed band and $0 \leq a_N < 1$. By (3.14),

$$C_N - R_N(a_N, b'_N) \geq R_N(0, b'_N) - R_N(a_N, b'_N) \geq d_{\text{loss}} a_N \frac{N^{-1/2}}{\log N}.$$

This yields the stated little-oh estimate on a_N . Moreover $0 \leq C_N - R_N(0, b'_N) \leq C_N - R_N(a_N, b'_N) = o(N^{-p})$. The compact profile convergence (3.18) and the strict maximum of f now imply $b'_N - \hat{b}_N \rightarrow 0$. $\square$

Remark 3.7 (What is and is not effective). The exponent, amplitude, center, and power error rate are explicit. The proof establishes an eventual axis theorem but does not supply a practical numerical value of N_0 : the range $s \geq 2 \log N$ uses the slowly decreasing factor $N^{1-2 \log(5/3)}$. One could change the splitting point from 2 to a number between 2 and the positive root of $\alpha - 1/2 - \alpha \log \alpha = 0$ to improve that technical convergence. No statement that the axis reduction holds for *every* index is proved here. That finite-index conjecture remains a separate target.

Remark 3.8 (Small outer orders and the mechanism). The incoming lower asymptotic alone suggested axis extremizers. The missing mechanism is the loss $J_N(a)$: it is at least a constant times $aN^{-1/2}/\log N$, whereas the possible gain from moving the coefficient 3^{-a} is only $O(aN^{-p})$. Since $p > 1/2$, every positive outer order gives a smaller error than the axis in the only inner-order band that can compete with the axis. The positivity and monotonicity of the exact Taylor remainder are what make this comparison rigorous without differentiating a big-oh term.

3.7 The sharp joint profile at the outer-order boundary

The necessary scale in Theorem 3.2 has a matching sufficiency statement and an explicit limiting loss. Define

$$s_N^{\text{out}} = N^{-(p-1/2)} \log N, \quad f(s) = q^{-p} 2^{-s} - q^{-(p+1)} 3^{-s}.$$

Theorem 3.9 (Joint boundary profile and a complete near-optimality criterion). *Uniformly for (ξ, s) in any compact subset of $[0, \infty) \times \mathbb{R}$,*

$$N^p \left(R_N(\xi s_N^{\text{out}}, \hat{b}_N + s) - 1 \right) \longrightarrow f(s) - \sqrt{\pi} \xi. \quad (3.19)$$

For any sequences $a_N \geq 0$, $b'_N > 0$,

$$C_N - R_N(a_N, b'_N) = o(N^{-p}) \iff \begin{cases} b'_N - \hat{b}_N \longrightarrow 0, \\ a_N = o(s_N^{\text{out}}). \end{cases} \quad (3.20)$$

Proof. It remains to compute the outer-order loss more sharply. Put $L = \frac{1}{2} \log N$. For every fixed $M > 0$, uniformly for $0 < a \leq M s_N^{\text{out}}$,

$$J_N(a) = \sqrt{\pi} a \frac{N^{-1/2}}{\log N} \{1 + o(1)\}. \quad (3.21)$$

To verify it, first restrict its defining gamma integral to $t \geq L/2$ and set $t = L + y$. After extracting $N^{-1/2} L^{a-1}/\Gamma(a)$, the remaining integral is

$$\int_{-L/2}^{\infty} (1 + y/L)^{a-1} e^{-y} W_N(N^{-1} e^{-2y}) dy.$$

For all large N , $a \leq 1/2$. The power factor is at most two when $-L/2 \leq y \leq 0$ and at most one for $y \geq 0$. Also $W_N(z) \leq e^{-Nz}$ for $0 \leq z \leq 1$. An integrable common majorant is therefore $2e^{-y} \exp(-e^{-2y})$ on the whole real line. The integrand converges, uniformly in the stated small- a range on each compact interval, to $e^{-y} \exp(-e^{-2y})$. Its integral is

$$\int_{-\infty}^{\infty} e^{-y} e^{-e^{-2y}} dy = \frac{1}{2} \Gamma(1/2) = \frac{\sqrt{\pi}}{2}.$$

The omitted interval is uniformly negligible, including as $a \downarrow 0$. For $0 < t \leq 1/2$,

$$W_N(e^{-2t}) \leq (2t)^{N/2} e^{-N/(2e)};$$

this follows by bounding one half of $(1 - e^{-2t})^N$ by $(2t)^{N/2}$ and the other by $e^{-(N/2)e^{-2t}}$. Its integral against $t^{a-1}e^{-t}$ is at most $2^{-a}e^{-N/(2e)}/(a + N/2)$. For $1/2 \leq t \leq L/2$, use $W_N(e^{-2t}) \leq e^{-\sqrt{N}}$ and $t^{a-1} \leq 2$, giving at most $Le^{-\sqrt{N}}$. Both bounds are negligible relative to $N^{-1/2}L^{a-1}$. Finally $\Gamma(a)^{-1}/a \rightarrow 1$ and $L^a \rightarrow 1$ uniformly in the indicated range, because $s_N^{\text{out}} \log L \rightarrow 0$. This proves (3.21); at $a = 0$ both sides vanish.

Subtract the exact expansion (3.9) at $a = 0$ from that at $a = \xi s_N^{\text{out}}$ and $b = \hat{b}_N + s$. The difference is

$$-\zeta(b)J_N(a) + (N+1)(1-3^{-a})S_2(b) + \eta_N(a, b) - \eta_N(0, b).$$

On a compact s interval, $\zeta(b) = 1 + o(1)$, $NS_2(b) = O(N^{-p})$, and $0 \leq \eta_N(a, b) \leq \eta_N(0, b) = O(N^{-p-\delta})$. The last two displayed terms are consequently $o(N^{-p})$ uniformly for bounded ξ . Equation (3.21) and the axis profile (3.18) prove (3.19). The forward implication of (3.20) is Theorem 3.2. For the reverse implication, use (3.19) with $\xi \rightarrow 0$, $s \rightarrow 0$, $f(0) = c$, and the already proved expansion of C_N . $\square$

4 A second asymptotic scale from the gamma transition region

The elementary Taylor envelope gives a power remainder and proves the matching leading constant. Its positive remainder has a more precise asymptotic. The relevant gamma mass is centered at the transition $t = \frac{1}{2} \log N$, rather than at either of the two moment saddles that produce 2^{-b} and $N3^{-b}$.

Retain

$$d = \log(3/2), \quad \lambda = d^{-1}, \quad p = \lambda \log 2, \quad q = \frac{\log 3}{\log 2}, \quad B_0 = \lambda \log q, \quad \hat{b}_N = \lambda \log N + B_0.$$

Since $q = 1 + 1/p$, the leading amplitude has the equivalent form

$$c = (1 - q^{-1})q^{-p} = \frac{p^p}{(p+1)^{p+1}}. \quad (4.1)$$

Introduce the additional constants

$$\alpha = \lambda - \frac{1}{2}, \quad \rho = 2\lambda - 1 = 2\alpha, \quad \mu = \frac{1}{2} - \lambda + \lambda \log(2\lambda), \quad (4.2)$$

$$D_0 = \frac{\Gamma(-\alpha)}{2} \sqrt{\frac{2\lambda}{\pi}} (2\lambda)^{-B_0}, \quad K = (\log 2)dq^{-p}, \quad (4.3)$$

$$D_1 = \frac{\log(2\lambda)D_0}{K}.$$

Here $1 < \alpha < 2$, so $\Gamma(-\alpha) > 0$. The exponents obey

$$p < \mu < 2p - 1. \quad (4.4)$$

For orientation, numerical evaluations are

$$\mu = 1.9695904144700607703\dots, \quad \mu - p = 0.2600791231186059934\dots,$$

$$D_0 = 1.5677702007103631657\dots, \quad D_1 = 19.562981709827216349\dots$$

The exact formulas, rather than these decimals, enter the proofs.

Theorem 4.1 (The next term and the direction of the maximizing shift). *Let b_N be the unique axis maximizer, which also gives the global constant C_N for every sufficiently large N . Then*

$$C_N = 1 + cN^{-p} + D_0 \frac{N^{-\mu}}{\sqrt{\log N}} + o\left(\frac{N^{-\mu}}{\sqrt{\log N}}\right), \quad (4.5)$$

$$b_N = \widehat{b}_N - D_1 \frac{N^{-(\mu-p)}}{\sqrt{\log N}} + o\left(\frac{N^{-(\mu-p)}}{\sqrt{\log N}}\right). \quad (4.6)$$

In particular the true maximizing inner order is eventually smaller than $\widehat{b}_N$, and the leading approximation $1 + cN^{-p}$ eventually underestimates the sharp constant.

4.1 The exact remainder and its transition limit

Put

$$G_N(z) = \frac{(1-z)^{N-1}}{1+z}, \quad r_N(z) = G_N(z) - 1 + Nz.$$

The earlier exact axis expansion reads

$$R_N(0, b) = 1 + 2^{-b} - N3^{-b} - N4^{-b} + \varepsilon_N(b), \quad (4.7)$$

where

$$\varepsilon_N(b) = \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} e^{-t} (1 + e^{-t}) r_N(e^{-2t}) dt. \quad (4.8)$$

The nonnegative Taylor remainder satisfies both bounds

$$0 \leq r_N(z) \leq \min\{Nz, A_N z^2\}, \quad A_N = (N^2 - N + 2)/2. \quad (4.9)$$

The quadratic bound is the previous Taylor theorem. The linear bound uses $0 \leq G_N(z) \leq 1$.

Lemma 4.2 (Uniform transition asymptotics with two derivatives). *Let $\ell = \log N$, let B range over any fixed compact real interval, and set $b = \lambda\ell + B$. Uniformly in B , for $j = 0, 1, 2$,*

$$\frac{d^j}{db^j} \varepsilon_N(b) = \frac{\Gamma(-\alpha)}{2} \sqrt{\frac{2\lambda}{\pi}} (2\lambda)^{-B} \{-\log(2\lambda)\}^j \frac{N^{-\mu}}{\sqrt{\ell}} + o\left(\frac{N^{-\mu}}{\sqrt{\ell}}\right). \quad (4.10)$$

Proof. Set $L = \ell/2$ and

$$P_N(B) = \frac{e^{-L} L^{b-1}}{\Gamma(b)}.$$

With $t = L + y$, equation (4.8) becomes

$$\frac{\varepsilon_N(b)}{P_N(B)} = \int_{-L}^\infty D_{N,B}(y) (1 + N^{-1/2} e^{-y}) r_N(N^{-1} e^{-2y}) dy, \quad (4.11)$$

where

$$D_{N,B}(y) = \exp\{(b-1) \log(1 + y/L) - y\}.$$

For each fixed y , uniformly for bounded B ,

$$D_{N,B}(y) \longrightarrow e^{\rho y}, \quad r_N(N^{-1} e^{-2y}) \longrightarrow e^{-e^{-2y}} - 1 + e^{-2y}.$$

The integrands vanish below $-L$.

Choose a fixed $\epsilon > 0$ so small that $2 < \rho - 3\epsilon < \rho + 3\epsilon < 4$. The inequality $\log(1 + u) \leq u$ and boundedness of B give, for all sufficiently large N ,

$$D_{N,B}(y) \leq \begin{cases} e^{(\rho+\epsilon)y}, & y \geq 0, \\ e^{(\rho-\epsilon)y}, & -L < y < 0. \end{cases}$$

Also $1 + N^{-1/2}e^{-y} \leq 2$ on this domain. Equation (4.9) bounds the remaining factor by $\min(e^{-2y}, Ce^{-4y})$, with a fixed C . The resulting dominating function is integrable at both infinities: use the quartic bound for $y \geq 0$ and the quadratic-in- e^{-y} bound for $y < 0$. Dominated convergence therefore gives

$$\frac{\varepsilon_N(b)}{P_N(B)} \longrightarrow I_\rho := \int_{-\infty}^{\infty} e^{\rho y} (e^{-e^{-2y}} - 1 + e^{-2y}) dy = \frac{1}{2} \Gamma(-\alpha). \quad (4.12)$$

To verify the last evaluation, set $x = e^{-2y}$:

$$I_\rho = \frac{1}{2} \int_0^{\infty} x^{-\alpha-1} (e^{-x} - 1 + x) dx.$$

Two integrations by parts have zero boundary terms because $1 < \alpha < 2$, and give $I_\rho = \Gamma(2 - \alpha)/(2\alpha(\alpha - 1)) = \Gamma(-\alpha)/2$.

For clarity, the same domination controls parameter derivatives; this is not an inference from differentiating an asymptotic formula. The first two derivatives of the gamma density insert the factors

$$S_{N,B}(y) = \log(L + y) - \psi(b), \quad S_{N,B}(y)^2 - \psi_1(b).$$

At each fixed y these tend uniformly to $-\log(2\lambda)$ and $\log^2(2\lambda)$, respectively. For $y \geq 0$, their absolute values are bounded by a fixed multiple of $1 + y^2$, which is integrable against the preceding exponential majorant after a harmless adjustment of ϵ . For negative y write $u = 1 + y/L \in (0, 1)$. For $j = 1, 2$ and large L ,

$$u^{b-1} |\log u|^j \leq C_j u^{b-1-\epsilon L},$$

because $\sup_{v \geq 0} v^j e^{-\epsilon L v}$ is uniformly bounded. Replacing $b - 1$ by $b - 1 - \epsilon L$ in the preceding exponential estimate still leaves an exponent strictly greater than two on the negative half-line. The terms $\log L - \psi(b)$ and $\psi_1(b)$ are uniformly bounded. This supplies a common integrable majorant for the differentiated integrands as well.

Convergence is uniform for bounded B : the pointwise convergences are uniform on each compact y interval and the same integrable majorants apply to every B in the prescribed compact interval. Finally, uniform Stirling asymptotics [12] at $b = \lambda\ell + B$ give

$$P_N(B) = \sqrt{\frac{2\lambda}{\pi}} (2\lambda)^{-B} \frac{N^{-\mu}}{\sqrt{\ell}} \{1 + O(\ell^{-1})\}. \quad (4.13)$$

Combining this with the three dominated-convergence limits proves (4.10). $\square$

4.2 Optimization at the second scale

Proof of Theorem 4.1. We first record exact inequalities between the exponents, to avoid using numerical approximations as premises. The elementary bounds $2/5 < \log(3/2) < 5/12$ give $12/5 < \lambda < 5/2$. The function $x \mapsto \frac{1}{2} - x + x \log x$ is increasing for $x > 1$, and is positive at $x = 12/5$; for the latter assertion, $\log(12/5) > 2(12/5 - 1)/(12/5 + 1) = 14/17$ suffices. It follows that $\mu - p > 0$. Also

$$\mu - (2p - 1) = \frac{3}{2} - \lambda + \lambda \log(\lambda/2) < 0 :$$

the expression is increasing on $2 < \lambda < 5/2$, and at $5/2$ it is $-1 + (5/2)\log(5/4) < -3/8$. This proves (4.4).

Define the two scales

$$h_N = \frac{N^{-\mu}}{\sqrt{\ell}}, \quad t_N = N^p h_N = \frac{N^{-(\mu-p)}}{\sqrt{\ell}}.$$

For bounded s , set

$$f(s) = q^{-p}2^{-s} - q^{-(p+1)}3^{-s}, \quad g(s) = D_0(2\lambda)^{-s}.$$

Equation (4.7), Lemma 4.2, and $\mu < 2p - 1$ give the expansion

$$R_N(0, \hat{b}_N + s) = 1 + N^{-p}f(s) + h_N g(s) + o(h_N), \quad (4.14)$$

uniformly with its first two derivatives in s on every compact interval. In fact the omitted term $N4^{-b}$ and both of its derivatives are $O(N^{-(2p-1)}) = o(h_N)$.

The already proved localization gives $s_N = b_N - \hat{b}_N \rightarrow 0$. Differentiating the integral through the justified (4.14), the stationary equation is

$$0 = f'(s_N) + t_N g'(s_N) + o(t_N).$$

Since $f'(0) = 0$ and $f''(0) = -K < 0$, this first implies $s_N = O(t_N)$, and then

$$-K s_N - t_N \log(2\lambda) D_0 + o(t_N) = 0.$$

This is (4.6). Finally $f(s_N) = c + O(t_N^2)$ and $g(s_N) = D_0 + O(t_N)$. The displacement changes the height by $O(N^{-p}t_N^2) = O(h_N t_N) = o(h_N)$. Substituting into (4.14) proves (4.5). $\square$

4.3 The first inverse-logarithmic correction

The convergence of the second-scale amplitude is particularly slow because α is close to two. The first correction describes this effect explicitly. Let ψ and ψ_1 denote the digamma and trigamma functions, continued to $-\alpha$, which is not a pole. Put

$$T(B) = -(B-1)\psi(-\alpha) - \frac{\lambda}{2}\{\psi_1(-\alpha) + \psi(-\alpha)^2\} - \frac{B^2 - B + 1/6}{2\lambda}. \quad (4.15)$$

Proposition 4.3 (An explicit logarithmic refinement). *Uniformly for bounded B ,*

$$\varepsilon_N(\lambda\ell + B) = \frac{\Gamma(-\alpha)}{2} \sqrt{\frac{2\lambda}{\pi}} (2\lambda)^{-B} \frac{N^{-\mu}}{\sqrt{\ell}} \left\{ 1 + \frac{T(B)}{\ell} + O(\ell^{-2}) \right\}. \quad (4.16)$$

Consequently the global constant has the refinement

$$C_N = 1 + cN^{-p} + D_0 \frac{N^{-\mu}}{\sqrt{\ell}} \left\{ 1 + \frac{T(B_0)}{\ell} + O(\ell^{-2}) \right\}. \quad (4.17)$$

Proof. All moments in y of the limiting integrand in (4.12) are finite, by the same two exponential tail bounds. On a central interval $|y| \leq M \log \ell$, Taylor expansion gives

$$D_{N,B}(y) = e^{\rho y} \left\{ 1 + \frac{(B-1)y - \lambda y^2}{L} + O\left(\frac{1 + |y|^6}{L^2}\right) \right\},$$

uniformly for bounded B ; the harmless constant in the remainder can be chosen uniformly for all sufficiently large ℓ . On the same interval the finite- N kernel can be replaced by its limit with an error smaller than every fixed negative power of ℓ . Indeed $e^{-2y} = O(\ell^{2M})$, $N^{-1}e^{-2y} \rightarrow 0$ faster than every power of $1/\ell$, and expanding $\log G_N(N^{-1}e^{-2y})$ around zero gives this assertion; the factor $1 + N^{-1/2}e^{-y}$ has the same property. For each fixed target power of $1/\ell$, choose M large enough that the two uniform exponential tail bounds from Lemma 4.2 make the integrals outside this central interval smaller than that power. Integrating the displayed Taylor expansion, and extending its polynomial moments to the entire real line, therefore gives

$$\frac{\varepsilon_N(\lambda\ell + B)}{P_N(B)} = I_\rho + \frac{(B-1)I'_\rho - \lambda I''_\rho}{L} + O(L^{-2}).$$

Here primes differentiate the explicit function $I_\rho = \Gamma(-\rho/2)/2$ in ρ ; hence

$$\frac{I'_\rho}{I_\rho} = -\frac{1}{2}\psi(-\alpha), \quad \frac{I''_\rho}{I_\rho} = \frac{1}{4}\{\psi_1(-\alpha) + \psi(-\alpha)^2\}.$$

The uniform Stirling expansion one order beyond (4.13) is

$$P_N(B) = \sqrt{\frac{2\lambda}{\pi}}(2\lambda)^{-B} \frac{N^{-\mu}}{\sqrt{\ell}} \left\{ 1 - \frac{B^2 - B + 1/6}{2\lambda\ell} + O(\ell^{-2}) \right\}.$$

Multiplication and $L = \ell/2$ give (4.16). For (4.17), the optimizing shift changes the height by $O(h_N t_N) = o(h_N \ell^{-2})$, and $N4^{-\widehat{b}_N} = o(h_N \ell^{-2})$ by the strict exponent inequality. Thus one can evaluate the refined remainder at $B = B_0$. $\square$

For orientation, $T(B_0) = -2098.4287350903144\dots$. This large coefficient explains why numerical indices such as 10^8 already illustrate the leading law well but do not yet closely approximate the second-scale amplitude D_0 . The theorem is an asymptotic statement as N tends to infinity; no practical onset index for the second expansion is asserted.

5 Strict log-convexity of the axis envelope

For real $t \geq 1$, define the continuous interpolation

$$\mathcal{A}(t) = \sup_{b>0} \frac{1}{\Gamma(b)} \int_0^\infty u^{b-1} e^{-u} \frac{(1+e^{-u})(1-e^{-2u})^{t-1}}{1+e^{-2u}} du.$$

At integer indices, $\mathcal{A}(N)$ is the optimized axis constant. For every real $t \geq 1$, the supremum is a finite maximum greater than one. To verify existence, note the endpoint limits in b : the limit at infinity is one, while the limit at zero is zero if $t > 1$ and one if $t = 1$. Also the kernel has expansion $1 + v - tv^2 + O_t(v^3)$ at $v = 0$. Its gamma average consequently equals $1 + 2^{-b} - t3^{-b} + O_t(4^{-b})$ as $b \rightarrow \infty$, and is greater than one for sufficiently large b . The last big-oh bound follows by extending the bounded Taylor remainder divided by v^3 continuously near zero and bounding it on the rest of $[0, 1]$.

Theorem 5.1 (Strict log-convex envelope). *For distinct real $s, t \geq 1$ and $0 < \theta < 1$,*

$$\mathcal{A}(\theta s + (1-\theta)t) < \mathcal{A}(s)^\theta \mathcal{A}(t)^{1-\theta}.$$

In particular, at every integer $N \geq 2$,

$$\mathcal{A}(N)^2 < \mathcal{A}(N-1)\mathcal{A}(N+1).$$

The globally optimized Euler constants satisfy $C_N^2 < C_{N-1}C_{N+1}$ for every sufficiently large N .

Proof. For fixed b , its axis integral is a positive moment of $(1 - e^{-2u})^{t-1}$. Strict Hölder inequality gives strict log-convexity in t , since $\log(1 - e^{-2u})$ is not constant under its positive gamma-weighted density. Choose a finite maximizing b for the intermediate argument $\theta s + (1 - \theta)t$. Apply strict Hölder at that one value of b , and bound its two endpoint integrals by their respective maxima. This yields the asserted strict inequality; it avoids the false general inference that an arbitrary supremum preserves *strict* convexity without an attained maximizer. The discrete conclusions follow by $\theta = 1/2$ and the eventual axis reduction. $\square$

6 The global limiting Euler-kernel extremum

The pointwise kernel optimum and the optimum after averaging over polylogarithm orders are different quantities. This section resolves the limiting-kernel conjecture in the incoming sharp-Euler report [4]. The argument also proves eventual uniqueness of the maximizer and gives a convergent expansion in the reciprocal truncation index.

For an integer $N \geq 1$, put

$$W_N(x) = \frac{(1-x)^N}{1+x}, \quad K_N(r, y) = \frac{W_N(ry^2) - W_N(r)}{1-y}, \quad M_N = \max_{(r,y) \in [0,1]^2} K_N(r, y). \quad (6.1)$$

At $y = 1$ the definition means $K_N(r, 1) = -2rW'_N(r)$. The divided difference is continuous on the closed square; hence the maximum exists. We will use $c = Nr$ for the rescaled first coordinate.

6.1 A decreasing envelope with uniform endpoint control

Lemma 6.1 (Exact power envelope). *For every integer $m \geq 2$, define*

$$B_m(y) = \frac{(1-y^2)^m}{(1-y)(1-y^{2m/(m-1)})^{m-1}}, \quad 0 < y < 1. \quad (6.2)$$

Its continuous endpoint values are

$$B_m(0) = 1, \quad B_m(1) = 2 \left(1 - \frac{1}{m}\right)^{m-1}.$$

Then

$$B_m(y) = \max_{0 \leq r \leq 1} \frac{(1-ry^2)^m - (1-r)^m}{1-y}. \quad (6.3)$$

The sequence B_m decreases pointwise on $[0, 1]$ to the continuous function

$$B(y) = (1+y)y^{2y^2/(1-y^2)}, \quad B(0) = 1, \quad B(1) = 2/e. \quad (6.4)$$

The convergence is uniform on $[0, 1]$.

Proof. For $v = y^2 \in (0, 1)$, differentiation in r shows that the unique maximum of $(1 - rv)^m - (1 - r)^m$ satisfies

$$\frac{1 - r}{1 - rv} = v^{1/(m-1)}.$$

Substitution proves (6.3) and (6.2). The endpoint $y = 1$ is also the direct maximum of $2mr(1 - r)^{m-1}$.

To prove monotonicity, set $q = 1/(m - 1)$ and $L(q) = \log(1 - v^{1+q})$. Then

$$\log B_m(y) = \log(1 + y) - \frac{L(q) - L(0)}{q}.$$

The function L is strictly concave, since

$$L''(q) = -\frac{(\log v)^2 v^{1+q}}{(1 - v^{1+q})^2} < 0.$$

Its secant slope from zero decreases as q increases. Thus decreasing q , which means increasing m , decreases $B_m(y)$. As $q \downarrow 0$, the secant slope tends to $-v \log v / (1 - v)$, giving (6.4). At $y = 1$ the same monotonicity follows from the monotone increase of $(1 + 1/k)^k$ to e , with $k = m - 1$; at zero all values are one. The limiting function is continuous at both endpoints because $y^2 \log y \rightarrow 0$ at zero and $2y^2 \log y / (1 - y^2) \rightarrow -1$ at one. Dini's theorem now gives uniform convergence on the compact interval. $\square$

Lemma 6.2 (Uniform domination and local scaling). *For $N \geq 2$,*

$$K_N(r, y) \leq B_N(y) \quad (0 \leq r, y \leq 1). \quad (6.5)$$

Moreover, on every compact rectangle $0 \leq c \leq A$, $0 \leq y \leq 1$,

$$K_N(c/N, y) \longrightarrow H(c, y) := \frac{e^{-cy^2} - e^{-c}}{1 - y}, \quad H(c, 1) = 2ce^{-c}, \quad (6.6)$$

uniformly, together with every fixed number of derivatives on compact sets where the variables are interior.

Proof. The positive geometric expansion

$$W_N(x) = \sum_{k \geq 0} 2^{-k-1} (1 - x)^{N+k}$$

converges uniformly on $[0, 1]$. Apply lemma 6.1 termwise and use $B_{N+k} \leq B_N$. This also proves the continuous endpoint case by passage to the limit.

For bounded z , the functions

$$h_N(z) = W_N(z/N) = \frac{(1 - z/N)^N}{1 + z/N}$$

converge in C^j to e^{-z} for every fixed j . At $y = 1$ use

$$\frac{h_N(cy^2) - h_N(c)}{1 - y} = -c(1 + y) \int_0^1 h'_N(cy^2 + tc(1 - y^2)) dt.$$

This removes the apparent singularity and proves the stated uniformity. $\square$

6.2 The unique limiting maximum

Theorem 6.3 (Global kernel limit and maximizer localization). *There is exactly one $y_\infty \in (0, 1)$ satisfying*

$$1 + y - y^2 - y^3 + 4y \log y = 0. \quad (6.7)$$

Put

$$c_\infty = \frac{-2 \log y_\infty}{1 - y_\infty^2}, \quad M_\infty = (1 + y_\infty) y_\infty^{2y_\infty^2/(1 - y_\infty^2)}. \quad (6.8)$$

Then

$$M_N \longrightarrow M_\infty > 1. \quad (6.9)$$

For every choice of global maximizers (r_N, y_N) ,

$$Nr_N \longrightarrow c_\infty, \quad y_N \longrightarrow y_\infty. \quad (6.10)$$

The limiting function H has a unique global maximum at (c_∞, y_∞) , with negative definite Hessian there.

Proof. For fixed $0 < y < 1$, direct differentiation shows that $H(c, y)$ has exactly one maximum in c , attained at $c(y) = -2 \log y / (1 - y^2)$; its value is $B(y)$. The logarithmic derivative of B has the sign of

$$F(y) = 1 + y - y^2 - y^3 + 4y \log y, \quad (\log B)'(y) = \frac{F(y)}{(1 - y^2)^2}.$$

We have $F(0+) = 1$, $F(1) = F'(1) = 0$, and

$$F''(y) = -\frac{2(3y - 2)(y + 1)}{y}.$$

Thus F' increases from $-\infty$ up to $y = 2/3$, and then decreases to zero. Its maximum is positive because $F'(2/3) = 7/3 + 4 \log(2/3) > 1/3$, using $\log(3/2) < 1/2$. It follows that F first decreases and then increases, with exactly one zero in $(0, 1)$. At that zero $F' < 0$, so $B''(y_\infty) < 0$. The limiting maximum exceeds one: at $y = 1/4$, $B(y) > 1$ is equivalent to $(5/4)^{15} > 16$.

The uniform envelope in lemma 6.2 gives $\limsup M_N \leq \max B = M_\infty$. Evaluating at $(r, y) = (c_\infty/N, y_\infty)$ gives the matching lower bound. If (r_N, y_N) maximizes K_N , then

$$M_N \leq B_N(y_N) = B(y_N) + o(1).$$

Uniqueness of the maximum of B and compactness force $y_N \rightarrow y_\infty$. In particular y_N stays away from zero and one. If $c_N = Nr_N$ were unbounded along a subsequence, the estimate

$$0 \leq K_N(r_N, y_N) \leq \frac{e^{-Nr_N y_N^2}}{1 - y_N}$$

would force its value to zero, a contradiction. Every bounded subsequence has a further limit c , and the local scaling theorem gives $H(c, y_\infty) = M_\infty$. The unique conditional maximum then gives $c = c_\infty$; this proves (6.10).

Finally $H_{cc} < 0$ at each conditional maximum. Differentiating $H(c(y), y) = B(y)$ shows that the Schur complement of H_{cc} in the full Hessian is $H_{yy} - H_{cy}^2/H_{cc} = B''(y_\infty) < 0$. This proves negative definiteness. $\square$

6.3 Convergent reciprocal-index expansions

Theorem 6.4 (Eventual uniqueness and analytic perturbation). *For all sufficiently large integers N , K_N has exactly one global maximizer. There are real analytic functions $c(s)$, $y(s)$, and $M(s)$ near $s = 0$ such that*

$$Nr_N = c(1/N), \quad y_N = y(1/N), \quad M_N = M(1/N).$$

Their Taylor series give convergent expansions in $1/N$ for large N . In particular,

$$M_N = M_\infty + \frac{\kappa_\infty}{N} + O(N^{-2}), \quad \kappa_\infty = \frac{c_\infty^2 y_\infty^2 (1 + y_\infty) e^{-c_\infty y_\infty^2}}{2} > 0. \quad (6.11)$$

Consequently $M_{N+1} < M_N$ for all sufficiently large N .

Proof. Define, with its removable value at $s = 0$,

$$\mathcal{W}(s, z) = \exp\left(\frac{\log(1 - sz)}{s} - \log(1 + sz)\right), \quad \mathcal{W}(0, z) = e^{-z},$$

and

$$\mathcal{K}(s, c, y) = \frac{\mathcal{W}(s, cy^2) - \mathcal{W}(s, c)}{1 - y}.$$

It is analytic near $(0, c_\infty, y_\infty)$, and at $s = 1/N$ it is $K_N(c/N, y)$. The analytic implicit function theorem applied to $(\mathcal{K}_c, \mathcal{K}_y)$, whose Jacobian is the invertible Hessian in theorem 6.3, gives a unique analytic stationary branch $(c(s), y(s))$. The localization theorem places every global maximizer in an arbitrarily small neighborhood of this point for large N . On a fixed smaller convex neighborhood, the Hessian remains negative definite for small s . Strict concavity there and the implicit-function uniqueness prove global uniqueness.

The first two terms are

$$\mathcal{K}(s, c, y) = H(c, y) + sH_1(c, y) + O(s^2),$$

where, writing $A(z) = z + z^2/2$,

$$H_1(c, y) = \frac{e^{-c}A(c) - e^{-cy^2}A(cy^2)}{1 - y}. \quad (6.12)$$

Stationarity gives $e^{-c_\infty} = y_\infty^2 e^{-c_\infty y_\infty^2}$. Substitution into (6.12) gives exactly κ_∞ in (6.11). The first location correction is the explicit matrix formula

$$\begin{pmatrix} c'(0) \\ y'(0) \end{pmatrix} = -(D^2 H(c_\infty, y_\infty))^{-1} \nabla H_1(c_\infty, y_\infty). \quad (6.13)$$

Higher corrections follow by ordinary power-series substitution. Because the optimized value is itself analytic and $M'(0) = \kappa_\infty > 0$, it increases with s near zero. Taking $s = 1/N$ proves eventual strict decrease. The two-term asymptotic alone, without this derivative information, would not justify the last conclusion. $\square$

The diagnostic values, computed independently of the proof, are

$$\begin{aligned} y_\infty &= 0.1453038592593584238735558631\dots, \\ c_\infty &= 3.9410648316472506681136817940\dots, \\ M_\infty &= 1.0538619303627116260728728105\dots, \\ \kappa_\infty &= 0.1727965939279004260875754084\dots \end{aligned}$$

The incoming conjecture concerning the limit and all maximizer sequences is therefore settled. The stronger assertion of uniqueness and decrease for *every* $N \geq 2$ remains a separate finite-index problem; section 7 resolves the uniqueness assertion at $N = 2$.

7 An exact algebraic solution at the first later truncation

The sharp-Euler incoming report explicitly proposed exact elimination and boundary comparison for K_2 as a first target [4, Section 8]. Both can be completed with small polynomials. This gives an exact answer, including uniqueness, rather than an enclosure of a numerically located stationary point.

Theorem 7.1 (The unique maximum of K_2). *Let y_2 be the unique root in $(0, 1)$ of*

$$R(y) = y^7 + 3y^6 - 2y^5 - 10y^4 + 17y^3 + 59y^2 - 4. \quad (7.1)$$

Define

$$D(y) = y^6 + 2y^5 - 2y^4 - 6y^3 + 17y^2 + 36y + 4, \quad (7.2)$$

$$r_2 = 12/D(y_2), \quad M_2 = K_2(r_2, y_2). \quad (7.3)$$

Then (r_2, y_2) is the unique global maximizer of K_2 on $[0, 1]^2$. The number M_2 has degree seven over $\mathbb{Q}$ and satisfies

$$\begin{aligned} 0 &= 32M_2^7 - 416M_2^6 + 2538M_2^5 - 4523M_2^4 \\ &\quad - 32216M_2^3 - 66352M_2^2 + 99072M_2 + 20736. \end{aligned} \quad (7.4)$$

In particular $M_2 = 1.11893868596105\dots$

Proof. The kernel simplifies to

$$K_2(r, y) = \frac{r(1+y)(3-r-ry^2-r^2y^2)}{(1+r)(1+ry^2)}. \quad (7.5)$$

Its partial derivatives, after removing positive denominators and the negative factors $-(1+y)$ and $-r$, have numerators

$$\begin{aligned} P(r, y) &= r^4y^4 + 2r^3y^4 + 2r^3y^2 + r^2y^4 + 8r^2y^2 \\ &\quad + r^2 + 2ry^2 + 2r - 3, \\ Q(r, y) &= r^3y^4 + r^2y^4 + 2r^2y^2 + 6ry^2 + 8ry + r - 3. \end{aligned}$$

The exact polynomial identities used in the elimination are

$$\text{Res}_r(P, Q) = -192y^{10}(y+1)R(y), \quad (7.6)$$

and the penultimate subresultant

$$16y^{10}(rD(y) - 12). \quad (7.7)$$

These are finite algebraic identities; the accompanying verifier reconstructs them from the derivative polynomials.

There is exactly one zero of R in $(0, 1)$. Indeed $R(0) = -4$, $R(1) = 64$, and

$$R'(y) = y(7y^5 + 18y^4 - 10y^3 - 40y^2 + 51y + 118) > 68y.$$

For each such y , $P(r, y)$ is strictly increasing in $r \geq 0$, and $P(0, y) = -3$, $P(1, y) = 4y^4 + 12y^2 > 0$. Thus at most one interior stationary point can occur.

We next prove that a global maximum is interior. At $r = 0$ the kernel is zero, and at $y = 0$ it is $1 - W_2(r) \leq 1$. At $y = 1$,

$$K_2(r, 1) = \frac{2r(1-r)(3+r)}{(1+r)^2} < 1,$$

because the denominator minus the numerator equals $(1-2r)^2 + r^2 + 2r^3 > 0$. At $r = 1$ and $0 < y < 1$, $\partial_p K_2 < 0$ by $P(1, y) > 0$, so moving into the square increases the value. Finally

$$K_2(3/4, 1/4) = \frac{8325}{7504} > 1.$$

Compactness therefore supplies an interior global maximum. It must be the unique stationary point identified above. The subresultant gives $r_2 = 12/D(y_2)$; its denominator is positive, since $D(y) \geq y^6 + 2y^5 + 9y^2 + 36y + 4$ on $(0, 1)$.

Substituting $r = 12/D(y)$ in (7.5), clearing its positive denominator, and eliminating y against $R(y)$ gives (7.4). To establish the degree assertion, reduce its primitive polynomial modulo 5 and make it monic. The exact certificate checks

$$\gcd(f, X^5 - X) = 1, \quad X^{5^7} \equiv X \pmod{f}.$$

Since 7 is prime, these are the finite-field irreducibility criterion in degree seven. Its leading coefficient does not vanish modulo 5, so Gauss's lemma proves irreducibility over $\mathbb{Q}$. Rational bisection of R followed by interval evaluation of (7.3) certifies the displayed decimal prefix. $\square$

The geometric significance of this computation is limited but precise. It solves a pointwise maximization problem for the remainder kernel. It does not identify the order-averaged C_2 : the two gamma shape parameters cannot be assumed to concentrate independently at (r_2, y_2) .

8 Diagonal Lerch velocities and their classical inverse series

The harmonic-order continuation report [2] defines, for integers $n \geq 0$, $k \geq 1$, lattice spacing $h > 0$, and $0 \leq \rho \leq 1$,

$$f_{n,k}(a) = (-1)^{n+k} \frac{d^k}{da^k} \left(\frac{\log^n a}{a} \right), \quad \mathcal{F}_{n,k}^{\rho,h}(a) = \sum_{m \geq 0} \rho^m f_{n,k}(a + mh).$$

For fixed n, k , the derivative series converges locally uniformly on $a > 0$, including at $\rho = 1$. When $k = n - 1$ and $n \geq 2$, the smallest positive elementary zero is $a = 1$, it is simple, and $f'_{n,n-1}(1) = -n!$. Consequently it has an analytic local continuation $a_n^{(h)}(\rho)$ near $\rho = 0$. Put $A = 1 + h > 1$ and

$$v_n(A) = -\frac{1}{n!} \frac{d^{n-1}}{da^{n-1}} \left(\frac{\log^n a}{a} \right) \Big|_{a=A}. \quad (8.1)$$

Implicit differentiation gives $(a_n^{(h)})'(0) = v_n(A)$ for $n \geq 2$. The auxiliary coefficient $v_1(A) = -(\log A)/A$ completes the generating series; no zero-branch interpretation is asserted for this auxiliary index.

Let $[n_j]$ denote the unsigned Stirling numbers of the first kind and write $L = \log A$. The baseline proves the exact identities

$$v_n(A) = A^{-n} \sum_{j=1}^n \left[\begin{matrix} n \\ n-j+1 \end{matrix} \right] \frac{(-L)^j}{j!}, \quad (8.2)$$

$$V_A(t) = \sum_{n \geq 1} v_n(A) t^n = L - U_A(t), \quad e^{U_A(t)} = A + tU_A(t), \quad U_A(0) = L. \quad (8.3)$$

For clarity, the second formula is initially a statement about analytic germs at zero. It follows from Lagrange inversion [9] applied to $X = A + t \log X$, with $U_A = \log X$; the analytic implicit function theorem applies since the derivative with respect to U at $(0, L)$ is $A \neq 0$.

The polynomial $p_n(L) = A^n v_n(A)$ is also, up to the explicit sign $L_n(L) = (-1)^n p_n(L)$, Cohen's Lambert polynomial [6, Proposition 4.8]. This identification connects the velocity problem to the real-root and extreme-root questions treated in Sections 13 and 14.

These coefficient polynomials have a classical antecedent that should be made explicit. Kalugin and Jeffrey [5, §2.1, Theorem 1] consider

$$1 - e^{-v} + \sigma v - \sigma \lambda = 0, \quad v(0) = 0.$$

The exact change of variables

$$\lambda = L, \quad \sigma = -t/A, \quad v = V_A(t) \quad (8.4)$$

identifies their fixed- λ series with (8.3). Their double series, specialized by $\tau = \lambda \sigma$, gives exactly (8.2). Their convergence criterion also gives the radius obtained below. Thus neither the coefficient array nor the radius formula is presented as a new discovery in the broader literature. The purpose here is to give a direct barrier proof with full sheet selection and a radial-slit continuation, thereby settle the precisely stated accessibility conjecture in [2], and derive its quantitative consequences for the Lerch zero velocities.

9 An exact barrier selects the dominant conjugate pair

For $A > 1$ define $\theta \in (0, \pi)$ by

$$1 - \theta \cot \theta + \log \left(\frac{\theta}{\sin \theta} \right) = \log A, \quad (9.1)$$

and set

$$a = 1 - \theta \cot \theta, \quad u_* = a + i\theta, \quad \tau = e^{u_*}, \quad R = e^a. \quad (9.2)$$

The left side of (9.1) increases from zero to infinity, since its derivative is

$$\theta \csc^2 \theta + \frac{1}{\theta} - 2 \cot \theta = \frac{(\theta - \sin \theta \cos \theta)^2 + \sin^4 \theta}{\theta \sin^2 \theta} > 0.$$

Hence θ exists and is unique. Moreover $a > 0$, since $\sin \theta - \theta \cos \theta > 0$ on $(0, \pi)$, and

$$\frac{A}{R} = \frac{\theta}{\sin \theta}, \quad (1 - u_*)\tau = A. \quad (9.3)$$

Equivalently, $u_* = 1 + W_0(-A/e + i0)$, using the upper boundary value of the principal Lambert branch [11]. The angle definition avoids a dependence on boundary-value conventions.

Theorem 9.1 (Selected inverse sheet and its exact singularities). *For every $A > 1$, the analytic germ U_A in (8.3) has radius of convergence exactly R . Its only singularities on $|t| = R$ are τ and $\bar{\tau}$. There exists $\epsilon > 0$ such that it has a single-valued holomorphic continuation to*

$$\{t : |t| < R + \epsilon\} \setminus (\{s\tau : 1 \leq s < 1 + \epsilon/R\} \cup \{s\bar{\tau} : 1 \leq s < 1 + \epsilon/R\}). \quad (9.4)$$

At the upper endpoint, writing $z = 1 - t/\tau$, the continued branch has a convergent expansion

$$U_A(t) = u_* + \chi z^{1/2} + bz + c_3 z^{3/2} + b_2 z^2 + O(z^{5/2}), \quad (9.5)$$

where

$$\chi = \sqrt{-2u_*}, \quad \Re \chi > 0, \quad \Im \chi < 0, \quad b = \frac{u_*}{3} - 1, \quad c_3 = \chi \left(\frac{1}{2} - \frac{u_*}{18} - \frac{1}{4u_*} \right). \quad (9.6)$$

Here $z^{1/2}$ is positive on the inward radial segment $t = s\tau$, $s < 1$. The lower-endpoint expansion is the complex conjugate.

Proof. Write $F_t(u) = e^u - A - tu$. We first construct a domain in the u -plane that contains exactly one root for every $|t| < R$. Put $d = 1 - a$ and consider the rectangle

$$D_M = \{u = x + iy : 0 < x < M, |y| < \theta\},$$

where $M > L$ will be chosen large. On its upper horizontal side, set $h = x - a$. Equations (9.3) give the exact factorization

$$\begin{aligned} |e^{x+i\theta} - A|^2 - R^2(x^2 + \theta^2) \\ &= R^2((e^h - d)^2 - (h + 1 - d)^2) \\ &= R^2(e^h - 1 - h)(e^h + 1 + h - 2d). \end{aligned} \quad (9.7)$$

The same identity holds on the lower side. For $x \geq 0$ we have $h \geq -a$, and the second factor is increasing in h ; its value at $h = -a$ is $e^{-a} - 1 + a > 0$. The first factor is nonnegative and vanishes only at $h = 0$. Therefore

$$|e^u - A| \geq R|u| \quad (\Im u = \pm\theta, \Re u \geq 0), \quad (9.8)$$

with equality only at u_* on the upper side and $\bar{u}_*$ on the lower side.

On the left side, for $0 < |y| \leq \theta$,

$$\frac{|e^{iy} - A|^2}{y^2} = \frac{(A - 1)^2}{y^2} + A \left(\frac{\sin(y/2)}{y/2} \right)^2. \quad (9.9)$$

As a function of positive $y \leq \theta < \pi$, both terms decrease and the first decreases strictly. Its minimum occurs at $y = \theta$, where (9.7) is strictly positive because $x = 0 \neq a$. Thus $|e^{iy} - A| > R|y|$ on the entire left side; at $y = 0$ this is simply $A - 1 > 0$. Choose M so large that

$$e^M - A > R\sqrt{M^2 + \theta^2}.$$

The right side then satisfies the same strict bound.

For $|t| < R$ the full boundary of D_M consequently satisfies $|tu| < |e^u - A|$. Rouché's theorem says that F_t and $e^u - A$ have the same number of zeros in D_M , counted with multiplicity. The latter has exactly the one simple zero L : the other zeros $L + 2\pi ik$ are excluded by $\theta < \pi$. Hence F_t

also has exactly one zero, and it is simple. The implicit-function branches glue by uniqueness, so U_A is holomorphic for $|t| < R$, with

$$0 < \Re U_A(t) < M, \quad |\Im U_A(t)| < \theta. \quad (9.10)$$

If $|t_0| = R$ and $t_0 \notin \{\tau, \bar{\tau}\}$, then F_{st_0} has no boundary zero for $0 \leq s \leq 1$. For $s < 1$ this follows from the strict bounds. At $s = 1$ a boundary zero could occur only at one of the two equality points in (9.8), and then (9.3) would force t_0 to be the corresponding excluded parameter. The argument-principle count is therefore one throughout the homotopy. At t_0 there is precisely one simple root inside D_M , and the implicit function theorem continues U_A through t_0 .

It remains to identify the branch at a critical endpoint. Set $t = \tau(1 - z)$ and $u = u_* + w$. The exact local equation is

$$e^w - 1 - w + z(u_* + w) = 0. \quad (9.11)$$

Since $u_* \neq 0$, its two local inverses are convergent power series in $z^{1/2}$, beginning $w = \pm \chi z^{1/2} + O(z)$ with the choice of χ in (9.6). For positive real z sufficiently small, the plus branch has imaginary part strictly less than θ and real part inside $(0, M)$; it lies in D_M . The minus branch lies above the upper horizontal side. Since $|\tau(1 - z)| < R$, uniqueness inside D_M identifies the plus branch with U_A . The coefficients b, c_3 follow by substituting the Puiseux series into (9.11). Complex conjugation settles the lower endpoint. The nonzero square-root coefficient makes both endpoints genuine singularities, and the radius is exactly R .

For completeness, the local continuations give the full slit domain in (9.4). Take disjoint small disks around the two critical endpoints, with the selected Puiseux branches cut along the outward radial rays. At each regular point on $|t| = R$, choose an implicit-function disk centered there and small enough to avoid both critical endpoints. Such a disk avoids both radial cuts: the point of an outward ray closest to its center is the corresponding endpoint. Two overlapping disks whose centers lie on $|t| = R$ have an overlap meeting $|t| < R$, as seen by taking a suitable point of the chord between their centers. Regular overlaps, including regular/critical ones, are connected and contain no cut, while the critical disks are disjoint. Thus all analytic branches agree on overlaps by the identity theorem. A finite cover of the circle by the underlying uncut disks has a positive collar width. Subtracting the radial rays from this collar proves (9.4) after decreasing ϵ . $\square$

The proof selects a bounded inverse sheet directly. In particular, it does not minimize the moduli of all formal critical values $-A/W_k(-A/e)$, which accumulate at zero on other sheets. That minimization would be invalid for the germ under consideration. The radius agrees with [5, Eq. (2.9)] under (8.4); the barrier supplies the additional continuation and branch information used below.

10 Unconditional oscillatory asymptotics to arbitrary order

Theorem 10.1 (Two terms and all orders). *Fix $A > 1$ and let the parameters be as in (9.2)–(9.6). Define*

$$\chi_1 = \chi \left(-\frac{3}{8} + \frac{u_*}{12} + \frac{3}{8u_*} \right). \quad (10.1)$$

Then

$$v_n(A) = \frac{R^{-n}}{\sqrt{\pi} n^{3/2}} \Re \left[e^{-in\theta} \left(\chi + \frac{\chi_1}{n} \right) \right] + O_A(R^{-n} n^{-7/2}). \quad (10.2)$$

In particular, with $\phi = -\arg \chi \in (\pi/4, \pi/2)$,

$$v_n(A) = \frac{|\chi|}{\sqrt{\pi} R^n n^{3/2}} \cos(n\theta + \phi) + O_A(R^{-n} n^{-5/2}). \quad (10.3)$$

These are additive estimates, including at indices where the displayed cosine is small. An expansion to any prescribed number of inverse powers of n follows from explicitly computable local coefficients.

Proof. At τ , the nonintegral terms of $V_A = L - U_A$ are $-\chi(1 - t/\tau)^{1/2} - c_3(1 - t/\tau)^{3/2}$, with conjugate terms at $\bar{\tau}$. The slit domain from Theorem 9.1 permits the standard finite-singularity transfer theorem [8, Chapter VI]. For a fixed nonintegral exponent α ,

$$[t^n](1 - t/\tau)^\alpha = \frac{\tau^{-n}n^{-\alpha-1}}{\Gamma(-\alpha)} \left(1 + \frac{\alpha(\alpha+1)}{2n} + O(n^{-2}) \right).$$

Using $\Gamma(-1/2) = -2\sqrt{\pi}$ and $\Gamma(-3/2) = 4\sqrt{\pi}/3$, the upper singularity contributes

$$\frac{\tau^{-n}}{2\sqrt{\pi}n^{3/2}} \left(\chi + \frac{3\chi/8 - 3c_3/2}{n} \right) + O_A(R^{-n}n^{-7/2}).$$

Adding its conjugate and substituting for c_3 gives (10.2).

For an explicit all-orders construction, put

$$K(w) = \frac{2u_*(e^w - 1 - w)}{w^2(u_* + w)}, \quad K(0) = 1.$$

Equation (9.11) becomes $w = \chi z^{1/2} K(w)^{-1/2}$. Lagrange inversion gives the convergent Puiseux coefficients

$$U_A(t) = u_* + \sum_{m \geq 1} c_m z^{m/2}, \quad c_m = \frac{\chi^m}{m} [w^{m-1}] K(w)^{-m/2}. \quad (10.4)$$

For any integer $J \geq 1$, transfer of the odd coefficients yields the particularly convenient form

$$v_n(A) = -2\Re \left[\tau^{-n} \sum_{j=0}^{J-1} c_{2j+1} (-1)^n \binom{j + \frac{1}{2}}{n} \right] + O_{A,J}(R^{-n}n^{-J-3/2}). \quad (10.5)$$

The next omitted nonintegral exponent is $J + 1/2$; the integer powers are locally analytic and do not contribute algebraic transfer terms. Expanding the exact binomial coefficients by gamma-ratio asymptotics gives the requested expansion in inverse powers of n . $\square$

For $A = 2$, the exact angle definition gives numerically

$$u_* = 0.469353635658274 \dots + 1.132672497604881 \dots i, \quad R = 1.598960348180173 \dots,$$

and

$$\chi = 0.869892622200723 \dots - 1.302083117729355 \dots i.$$

The two-term formula in [2] is therefore now unconditional. These decimals merely illustrate the exact parameters; they have no role in the continuation proof.

11 Bounded sign gaps and a complete density alternative

The earlier infinite-sign theorem leaves open the lengths of possible runs in one direction. The singularity analysis gives a uniform bound on such runs for each fixed lattice spacing.

Corollary 11.1 (Every sufficiently late short block has both signs). *For $A > 1$ put*

$$K_A = \left\lfloor \frac{\pi}{\theta} \right\rfloor + 2.$$

Every sufficiently late block of K_A consecutive coefficients $v_n(A)$ contains a strictly positive and a strictly negative coefficient. In particular, every sufficiently late block of four coefficients $v_n(2)$ has both signs.

Proof. Put $m = K_A - 1$. Then $\pi < m\theta \leq \pi + \theta < 2\pi$. The points $0, \theta, \dots, m\theta$ on the circle have largest circular gap

$$g = \max\{\theta, 2\pi - m\theta\} < \pi.$$

Choose $\eta > 0$ with $\pi - 2\eta > g$. Every rotation of these points meets both an arc where $\cos x \geq \sin \eta$ and an arc where $\cos x \leq -\sin \eta$, since each arc has length $\pi - 2\eta$. After division by the positive prefactor in (10.3), the error is $O_A(1/n)$ and is eventually smaller than $\sin \eta/2$. Thus both strict signs occur.

For the final assertion, $\pi/3 < \theta(2) < \pi/2$ can be proved without decimal computation. At $\theta = \pi/3$ put $x = \pi/(3\sqrt{3}) \in (0, 1)$. The corresponding value of A is $2xe^{1-x} < 2$, by $\log x < x - 1$. At $\theta = \pi/2$ it is $\pi e/2 > 2$. Monotonicity in (9.1) gives the claimed bracket and hence $K_2 = 4$. $\square$

Theorem 11.2 (Sign densities for every lattice parameter). *If $\theta/(2\pi)$ is irrational, the positive and negative velocities each have natural density $1/2$, and the zero velocities have density zero. If $\theta/(2\pi) = p/q$ is rational in lowest terms, the velocity signs are eventually periodic of period dividing q , none is eventually zero, and their densities are*

$$d_+(A) = \frac{1}{q} \#\{0 \leq j < q : \cos(2\pi j/q + \phi) > 0\}, \quad d_-(A) = 1 - d_+(A). \quad (11.1)$$

For even q both densities are $1/2$; for odd q the positive count is $(q-1)/2$ or $(q+1)/2$.

Proof. In the irrational case the sequence $n\theta + \phi$ is equidistributed modulo 2π by Weyl's theorem [13]. For any $\delta > 0$, the additive estimate (10.3) implies that any disagreement with the cosine sign, including a zero coefficient, eventually occurs only where $|\cos(n\theta + \phi)| < \delta$. The density of this set tends to zero with δ . The asserted densities follow.

In the rational case it is essential to rule out a zero among the finite leading cosine values. Such a zero would make ϕ/π rational. Since $\arg u_* = \pi - 2\phi$, the positive number $b_0 = \theta/a = \tan(\arg u_*)$ would then be algebraic. Also $\cot \theta$ is algebraic when θ/π is rational. But $a = 1 - \theta \cot \theta$ implies

$$\theta(1 + b_0 \cot \theta) = b_0.$$

The coefficient of θ cannot vanish, because $b_0 > 0$. This would make θ algebraic, contradicting the transcendence of π [14]. Thus every leading cosine has a nonzero margin from zero, and the $O_A(1/n)$ error gives eventual periodicity. Multiplication by p permutes the q phase points, proving (11.1). Antipodal pairing gives the even- q statement. For odd q , an open semicircle avoiding the sample points contains $(q-1)/2$ or $(q+1)/2$ of the equally spaced points. $\square$

The rational-angle parameters form the explicit countable set

$$A = \frac{\theta}{\sin \theta} \exp(1 - \theta \cot \theta), \quad \theta = 2\pi p/q \in (0, \pi).$$

No irrationality claim is made for the particular angle associated with $A = 2$. The four-term sign-block result is unconditional and does not require deciding that arithmetic question.

12 Convergent identities at regular and critical parameters

The preceding estimates turn formal substitutions in the inverse equation into justified convergent identities. They also describe the first differentiated series at the critical endpoints.

Proposition 12.1 (Interior sums and critical boundary values). *For every $A > 1$, both of the following series converge absolutely:*

$$\sum_{n \geq 1} v_n(A) = \log A + A + W_{-1}(-e^{-A}), \quad (12.1)$$

$$\sum_{n \geq 1} (-1)^n v_n(A) = \log A - A + W_0(e^A). \quad (12.2)$$

On the critical circle the series still converges absolutely, and

$$\sum_{n \geq 1} v_n(A) R^n \cos(n\theta) = \log A - a, \quad (12.3)$$

$$\sum_{n \geq 1} v_n(A) R^n \sin(n\theta) = -\theta. \quad (12.4)$$

The differentiated critical series has the precise finite-part limit

$$\sum_{n=1}^N n v_n(A) \tau^n = \frac{\chi}{\sqrt{\pi}} \sqrt{N} + \frac{u_*}{3} - 1 + O_A(N^{-1/2}). \quad (12.5)$$

Proof. Since $a > 0$, we have $R > 1$, so substitution of $t = \pm 1$ lies strictly inside the disk of convergence. On the positive real U axis the solutions at those parameters are

$$U_A(1) = -A - W_{-1}(-e^{-A}), \quad U_A(-1) = A - W_0(e^A).$$

They are positive and satisfy the implicit equation. They are the selected roots by uniqueness in the rectangle, or by continuation along the real interval, where $U_A > 0$. Substitution into $V_A = \log A - U_A$ proves (12.1)–(12.2).

Equation (10.3) gives $\sum_{n \geq 1} |v_n(A)| R^n < \infty$. Abel continuity at τ and Theorem 9.1 therefore yield $\sum_{n \geq 1} v_n(A) \tau^n = \log A - u_*$. Real and imaginary parts give (12.3)–(12.4).

Finally write $r = t/\tau$ and $B(r) = r dV_A(\tau r)/dr = \sum_{n \geq 1} n v_n(A) \tau^n r^n$. The expansion at $r = 1$ is

$$B(r) = \frac{\chi}{2} (1-r)^{-1/2} + b + O((1-r)^{1/2}).$$

The partial sum in (12.5) is $[r^N]B(r)/(1-r)$. Its singular terms at $r = 1$ give

$$\frac{\chi}{2} [r^N] (1-r)^{-3/2} + b = \frac{\chi}{\sqrt{\pi}} \sqrt{N} + b + O_A(N^{-1/2}).$$

The other singularity is at $r = \bar{\tau}/\tau = e^{-2i\theta} \neq 1$. There the factor $1/(1-r)$ is regular and B has exponent $-1/2$, so its coefficient contribution is $O_A(N^{-1/2})$. Transfer of the remaining terms has the same or smaller order. This proves the finite-part formula. $\square$

All sums in this proposition become explicit Stirling identities on using (8.2). Their derivation is a consequence of the proved continuation and classical inverse representation; no claim of independent priority for each specialized identity is made.

13 The global distribution of zero-velocity lattice parameters

The same singularity geometry has a second interpretation: one may fix the index and vary the lattice spacing. Define

$$p_n(L) = \sum_{j=1}^n \left[\begin{matrix} n \\ n-j+1 \end{matrix} \right] \frac{(-L)^j}{j!}, \quad v_n(A) = A^{-n} p_n(\log A).$$

For $n \geq 2$, write the positive zeros of p_n in increasing order as $L_{n,1} < \cdots < L_{n,n-1}$, and put $A_{n,j} = e^{L_{n,j}}$. Thus $A_{n,j} - 1$ is a lattice spacing where the initial diagonal velocity vanishes. Equivalently, these $L_{n,j}$ are the positive zeros of Cohen's Lambert polynomial $L_n(L) = (-1)^n p_n(L)$ [6, Proposition 4.8]; the law below is therefore a bulk zero law for that known family.

Lemma 13.1 (Real-rootedness and compact uniformity). *The polynomial p_n has exactly the simple zero 0 and $n - 1$ simple positive zeros. The additive asymptotic (10.3) is uniform for A in any compact subinterval of $(1, \infty)$.*

Proof. The real-rootedness is the elementary diagonal result from [2]; here is its operator proof. With $D = d/dx$,

$$np_n(L) = \left(\prod_{j=1}^{n-1} (1 + D/j) \right) x^n \Big|_{x=-L}.$$

For a polynomial P with real roots r_i of multiplicities m_i , the roots of $P + cP'$ away from inherited roots solve $\sum_i m_i/(x - r_i) = -1/c$. For $c > 0$, the left side is strictly decreasing between its poles. There is one new simple root in each interval between distinct roots and one to the left of the smallest root; an inherited root loses one multiplicity. Starting with x^n and applying $n - 1$ such operators gives n simple nonpositive roots, of which zero is the largest. Replacing x by $-L$ proves the assertion.

For uniformity, fix a compact parameter interval. Its angles stay a positive distance from 0 and π , its values of a stay positive, and the two critical points remain separated. The critical parameters and local coefficients depend real analytically on A . In the normalized variable $r = t/R(A)$, the rectangle estimates used in Theorem 9.1 have uniform strict margins away from small neighborhoods of the two equality points. The right boundary can be chosen with one common sufficiently large M . On the compact set of regular boundary parameter pairs, the analytic implicit function theorem therefore supplies a uniform collar after taking a finite cover. At the two moving endpoints the normalized local equation (9.11) has a quadratic coefficient bounded away from zero, giving uniform Puiseux neighborhoods and remainder bounds, again by compactness. The resulting transfer contours may be chosen with fixed normalized sizes and fixed angular dents. The proof of Theorem 10.1 then gives a remainder constant independent of A in the chosen interval. $\square$

Theorem 13.2 (Limiting zero law and explicit bulk quantiles). *For every $A > 1$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n-1} \# \{j : A_{n,j} \leq A\} = \frac{\theta(A)}{\pi}. \quad (13.1)$$

Consequently, if $j_n/(n-1) \rightarrow x \in (0, 1)$, then

$$A_{n,j_n} \longrightarrow \frac{\pi x}{\sin(\pi x)} \exp(1 - \pi x \cot(\pi x)). \quad (13.2)$$

In the logarithmic coordinate $L = \log A$, the limiting probability measure has density

$$\frac{d\mu}{dL} = \frac{\theta}{\pi(a^2 + \theta^2)}, \quad L = 1 - \theta \cot \theta + \log \frac{\theta}{\sin \theta}, \quad a = 1 - \theta \cot \theta. \quad (13.3)$$

Equivalently, in the original coordinate $A > 1$ its density is

$$\frac{d\mu}{dA} = \frac{\theta(A)}{\pi A(a(A)^2 + \theta(A)^2)}. \quad (13.4)$$

Proof. Let $0 < \alpha < \beta < \pi$. The parameter A is a strictly increasing function of θ . Uniformly on $[\alpha, \beta]$, the polynomial $p_n(L(\theta))$ has the sign of

$$\cos(n\theta + \phi(\theta)) + O_{\alpha,\beta}(1/n),$$

apart from the ambiguity allowed by a small cosine. The smooth function ϕ has bounded derivative there, so $n\theta + \phi(\theta)$ is strictly increasing for all sufficiently large n . At its successive integer multiples of π , the leading cosines are alternately 1 and -1 , and the error is eventually smaller than $1/2$. Between successive such points the intermediate value theorem supplies a polynomial zero. The number of these successive pairs is $n(\beta - \alpha)/\pi + O_{\alpha,\beta}(1)$. Hence the lower limit of the fraction of zeros whose angle lies in $[\alpha, \beta]$ is at least $(\beta - \alpha)/\pi$.

Fix $\theta_0 \in (0, \pi)$. Applying this lower bound on $[\epsilon, \theta_0]$ gives a lower limit at least $(\theta_0 - \epsilon)/\pi$ for the fraction below θ_0 . Applying it on $[\theta_0 + \epsilon, \pi - \epsilon]$ and subtracting from the total of exactly $n - 1$ positive roots gives an upper limit at most $(\theta_0 + 2\epsilon)/\pi$. Letting ϵ decrease to zero proves (13.1). Strict monotonicity and continuity of its limiting distribution function give the quantile limit (13.2) by bracketing x between nearby continuity points.

Finally,

$$\frac{dL}{d\theta} = \theta \csc^2 \theta + \frac{1}{\theta} - 2 \cot \theta = \frac{(1 - \theta \cot \theta)^2 + \theta^2}{\theta} = \frac{a^2 + \theta^2}{\theta}.$$

Pushing forward the uniform measure $d\theta/\pi$ proves (13.3). Since θ runs from 0 to π , the density has total mass one, and the CDF also proves tightness of the empirical measures. $\square$

This distribution has a heavy logarithmic tail. As $L \rightarrow \infty$, $a \sim L$ and $\theta \rightarrow \pi$, so $d\mu/dL \sim L^{-2}$ and $\mu((L, \infty)) \sim L^{-1}$. There is no contradiction with the exact finite-degree first moment

$$\sum_{j=1}^{n-1} L_{n,j} = nH_{n-1}, \quad (13.5)$$

obtained by comparing the leading two coefficients of p_n : the limiting distribution has infinite mean. The bulk law concerns indices proportional to n ; it gives no uniform estimate for either extreme root, whose scaling requires separate analysis.

14 The extreme roots of Cohen's Lambert polynomials

The diagonal polynomials are a previously studied family. In the notation of Cohen [6, Proposition 4.8],

$$L_n(X) = (-1)^n p_n(X). \quad (14.1)$$

Cohen's Conjecture 7.1 concerns their real roots, interlacing, and extreme-root asymptotics, including a numerical constant in the largest root. We prove a full fixed-index extreme-root expansion, establish the conjectured polynomial dependence of every coefficient on the integer parameter, and identify the first constant. The proof also applies to Cohen's generalized polynomials $L_{n,k}$ for each fixed positive integer k . All references to that conjecture below concern the version dated 18 January 2021; no claim is made that every aspect of the conjecture remains unresolved elsewhere.

14.1 An operator normalization and a reciprocal transform

Let $n, k \geq 1$, set $N = n + k - 1$, and put $D = d/dx$. Cohen's signed-Stirling formula is equivalently

$$L_{n,k}(X) = \frac{(-1)^{nk}}{n} \left(\prod_{r=1}^{n-1} (1 + D/r) \right) x^N \Big|_{x=-X}. \quad (14.2)$$

An empty product is one. For $k = 1$, this gives (14.1). The operator argument in Lemma 13.1 proves immediately that $L_{n,k}$ has a zero of multiplicity k at the origin and exactly $n - 1$ additional zeros, all simple and positive. Indeed each application of $1 + cD$, $c > 0$, removes one multiplicity at zero and introduces one negative simple root, with the other negative roots strictly interlacing those of the preceding polynomial. Thus this elementary argument proves the real-rooted portions of Cohen's Conjecture 7.1(1),(4).

Proposition 14.1 (Interlacing of successive Lambert polynomials). *The zeros of L_n and L_{n+1} interlace. Their only common zero is 0, and all interlacing inequalities between positive zeros are strict.*

Proof. Write $T_n = \prod_{r=1}^{n-1} (1 + D/r)$ and $P_n(x) = T_n x^n$. For every real t ,

$$P_{n+1}(x) + tP_n(x) = T_n [x^n(x + (n+1)/n + t)]$$

is real-rooted, because its input polynomial is real-rooted and each factor $1 + D/r$ preserves that property. The elementary real-rooted pencil criterion implies weak interlacing. To recall why that criterion applies, cancel common factors in R/P , where every $R + tP$ is real-rooted. This real rational function cannot take a real value in the upper half-plane. Its imaginary part therefore has constant sign there; the monic degree difference one makes that sign positive. Its residues at the remaining real poles are negative, which forces strict interlacing of the remaining zeros and poles. Reintroducing common factors gives weak interlacing.

In the present case no nonzero root can be shared. The identity

$$P'_{n+1} = (n+1)(P_n + P'_n/n)$$

would make a shared nonzero root a multiple root of $P_{n+1} - (n+1)P_n/n = T_n x^{n+1}$. The operator argument proves that this last polynomial has only simple nonzero roots, with zero of multiplicity two, a contradiction. Changing x to $-X$ proves the assertion. This also proves the interlacing assertion in Cohen's Conjecture 7.1(3). $\square$

For $1 \leq j \leq n - 1$, write $X_{n,k,j}$ for the positive roots in *decreasing* order. This indexing is intentionally opposite to the bulk indexing in Section 13. Let $e_r = e_r(1, 1/2, \dots, 1/(n-1))$ be elementary symmetric functions, with $e_0 = 1$, and use $N^{\underline{r}} = N(N-1) \cdots (N-r+1)$. The reciprocal polynomial

$$Q_{n,N}(w) = \sum_{r=0}^{n-1} (-1)^r \frac{N^{\underline{r}}}{N^r} e_r w^r \quad (14.3)$$

has its $n - 1$ roots at $w = N/X_{n,k,j}$, in increasing order. It is obtained from $L_{n,k}(N/w)$ by multiplication by $n(w/N)^N / ((-1)^{k-1} k)$, so the transform introduces no finite nonzero roots. Define also

$$f_n(w) = \prod_{r=1}^{n-1} (1 - w/r) = \frac{\Gamma(n-w)}{\Gamma(n)\Gamma(1-w)}, \quad g(w) = \frac{1}{\Gamma(1-w)}. \quad (14.4)$$

At the apparent poles the gamma quotient is understood through the polynomial product. Coefficient extraction gives the exact identity

$$Q_{n,N}(w) = \frac{N!}{N^N} [z^N] e^{Nz} f_n(wz). \quad (14.5)$$

14.2 An exact logarithmic shift and a uniform saddle expansion

The important simplification is to remove $\log n$ *before* expanding a zero. Set $\ell = \log n$ and define

$$h_n(v) = n^v f_n(v), \quad T_{n,N}(v) = \frac{N!}{N^N} [z^N] e^{Nz} h_n(vz). \quad (14.6)$$

Combining exponentials in (14.5) and rescaling its coefficient variable gives the exact identity

$$Q_{n,N}(w) = r^N T_{n,N}(v), \quad r = 1 - \frac{w\ell}{N}, \quad v = \frac{w}{r}. \quad (14.7)$$

Indeed $e^{Nz} f_n(wz) = e^{(N-w\ell)z} h_n(wz)$, and replacing z by z/r proves the formula. At a root corresponding to $X = N/w$, the new coordinate is simply

$$v = \frac{N}{X - \log n}, \quad X = \log n + \frac{N}{v}. \quad (14.8)$$

All these transformations are used near a fixed positive v , where r is nonzero for large n . Zeros far from this region need not be classified to determine any fixed number of largest roots.

The uniform gamma-ratio expansion [12, §5.11] gives, locally uniformly in the complex plane and with every fixed number of derivatives,

$$h_n(v) \sim g(v) R(v; 1/n), \quad R(v; \epsilon) = \exp \left(\sum_{a \geq 1} \lambda_a(v) \epsilon^a \right), \quad (14.9)$$

where

$$\lambda_a(v) = \frac{(-1)^{a+1}}{a(a+1)} (B_{a+1}(-v) - B_{a+1}(0)). \quad (14.10)$$

Here $B_m(x)$ are the Bernoulli polynomials defined by $te^{xt}/(e^t - 1) = \sum_{m \geq 0} B_m(x) t^m / m!$. The symbol $\sim$ denotes an asymptotic expansion to every finite order; no convergence of the formal series $R(v; \epsilon)$ is assumed. Its first coefficient is

$$R(v; \epsilon) = 1 + q_1(v) \epsilon + O(\epsilon^2), \quad q_1(v) = \frac{v(v+1)}{2}. \quad (14.11)$$

In particular h_n and each of its fixed derivatives are uniformly bounded on every fixed compact set. The gamma quotient is analytic on a fixed compact set for large n , and its expansion there follows from uniform Stirling expansion; derivative estimates follow by Cauchy's formula on a slightly larger compact set.

Lemma 14.2 (An all-orders coefficient-extraction estimate). *For $r \geq 0$, define polynomials $m_r(\eta)$ by*

$$\sum_{r \geq 0} m_r(\eta) \frac{s^r}{r!} = \exp \left(\sum_{q \geq 2} \frac{(-1)^{q-1}}{q} s^q \eta^{q-1} \right), \quad (14.12)$$

and differential operators

$$\mathcal{D}_b F(v) = \sum_{r=0}^{2b} [\eta^b] m_r(\eta) \frac{v^r}{r!} F^{(r)}(v). \quad (14.13)$$

Then, for each fixed integer $M \geq 0$ and compact $K \subset \mathbb{C}$,

$$T_{n,N}(v) = \sum_{b=0}^M N^{-b} \mathcal{D}_b h_n(v) + O_{K,M}(N^{-M-1}) \quad (14.14)$$

uniformly for $v \in K$ and all sufficiently large independent positive integers n, N . In particular,

$$\begin{aligned}\mathcal{D}_0 F &= F, & \mathcal{D}_1 F &= -\frac{v^2}{2} F'', \\ \mathcal{D}_2 F &= \frac{v^3}{3} F''' + \frac{v^4}{8} F^{(4)}, \\ \mathcal{D}_3 F &= -\frac{v^4}{4} F^{(4)} - \frac{v^5}{6} F^{(5)} - \frac{v^6}{48} F^{(6)}.\end{aligned}\tag{14.15}$$

Consequently $T_{n,N} \rightarrow g$ locally uniformly as $n, N \rightarrow \infty$.

Proof. On $z = e^{i\varphi}$ the exact coefficient integral is

$$\frac{N!}{2\pi N^N} \int_{-\pi}^{\pi} e^{Ne^{i\varphi} - iN\varphi} h_n(v e^{i\varphi}) d\varphi.$$

After extracting e^N , Stirling's formula bounds its prefactor by $O(\sqrt{N})$, while the modulus of its exponential factor is $e^{N(\cos \varphi - 1)}$. On $|\varphi| \leq N^{-2/5}$, expand the amplitude about $z = 1$ through degree $2M+1$. The uniform compact bounds on $h_n^{(2M+2)}$ bound the Taylor remainder by $C_{K,M}|e^{i\varphi} - 1|^{2M+2}$, independently of large n . Its integral is $O_{K,M}(N^{-M-1})$, since

$$\sqrt{N} \int e^{-cN\varphi^2} |\varphi|^{2M+2} d\varphi = O_M(N^{-M-1}).$$

On the complementary arc the exponential is at most $e^{-cN^{1/5}}$. The amplitude and its finite Taylor polynomial remain uniformly bounded on a fixed compact set, so this arc is negligible. We may therefore integrate the Taylor polynomial over the whole circle.

Its exact moments are

$$M_r(N) = \frac{N!}{N^N} [z^N] e^{Nz} (z-1)^r = \sum_{a=0}^r (-1)^{r-a} \binom{r}{a} \frac{N^a}{N^a}.$$

Their exponential generating function is exactly

$$\sum_{r \geq 0} M_r(N) \frac{s^r}{r!} = e^{-s} (1 + s/N)^N = \exp \left(\sum_{q \geq 2} \frac{(-1)^{q-1} s^q}{q N^{q-1}} \right).$$

Thus $M_r(N) = m_r(1/N)$. Moreover $m_0 = 1$, $m_1 = 0$, and for $r \geq 2$ the least possible power of η is $\lceil r/2 \rceil$: a product of terms $s^q \eta^{q-1}$ has s -degree at most twice its η -degree. This proves both the finite range in (14.13) and the fact that the Taylor term of degree $2M+1$ contributes only $O(N^{-M-1})$. Grouping the other terms by powers of $1/N$ proves (14.14). The displayed operators follow directly from the generating function. $\square$

14.3 All-orders edge asymptotics and polynomial parameter dependence

Write $H_m^{(r)} = \sum_{a=1}^m a^{-r}$, $H_0^{(r)} = 0$, and $H_m = H_m^{(1)}$. Let γ be Euler's constant.

Theorem 14.3 (All-orders fixed-index largest roots). *Fix positive integers j, k . There are explicitly computable polynomials $C_{j,m}(\kappa)$, $m \geq 1$, with $\deg C_{j,m} = m-1$, such that for every fixed integer $M \geq 0$,*

$$X_{n,k,j} = \frac{n+k-1}{j} + \log n + \gamma - H_{j-1} + \sum_{m=1}^M \frac{C_{j,m}(k-1)}{n^m} + O_{j,k,M}(n^{-M-1}).\tag{14.16}$$

The sum is empty when $M = 0$. Put

$$s_2 = H_{j-1}^{(2)} + \zeta(2), \quad s_3 = H_{j-1}^{(3)} - \zeta(3).$$

The first coefficient, independent of κ , is

$$C_{j,1}(\kappa) = j^2 s_3 - j s_2 - j - \frac{1}{2} =: C_j. \quad (14.17)$$

For $m \geq 2$, the coefficient of κ^{m-1} in $C_{j,m}(\kappa)$ is

$$(-1)^{m-1}(j^2 s_3 - j s_2), \quad (14.18)$$

which is nonzero. In particular, for every fixed $k \geq 1$,

$$X_{n,k,1} = n + k - 1 + \log n + \gamma - \frac{\frac{3}{2} + \zeta(2) + \zeta(3)}{n} + O_k(n^{-2}). \quad (14.19)$$

Proof. First identify the relevant roots. The limit in Lemma 14.2 is $g(v) = 1/\Gamma(1-v)$, whose zeros are the positive integers, all simple. Rouché's theorem on small disjoint circles about $1, \dots, j$ and uniform nonvanishing on the intervening compact real intervals, including the interval starting at 0, show that $T_{n,N}$ has one root near each of $1, \dots, j$ and no additional smaller positive roots. The nearby roots are real, either by conjugation and uniqueness in each circle or by the real-rootedness already established for the original polynomials. For large n , (14.7) maps each of these roots to a positive root $X = \log n + N/v$. Positive v correspond precisely to roots with $X > \log n$ and preserve decreasing order in X . Roots with $v < 0$ give $X < \log n$ and cannot intervene among these largest roots. Hence the root approaching j is exactly

$$v_{n,k,j} = \frac{N}{X_{n,k,j} - \log n}.$$

We now use formal series only as a finite-order computational device. Introduce independent formal variables ϵ, η and define

$$\mathcal{T}(v; \epsilon, \eta) = \sum_{b \geq 0} \eta^b \mathcal{D}_b(g(v)R(v; \epsilon)). \quad (14.20)$$

The gamma expansion and Lemma 14.2 show that this is the full two-parameter asymptotic expansion of $T_{n,N}$ when $\epsilon = 1/n$ and $\eta = 1/N$. More precisely, its truncation through total degree d has error $O_{K,d}((1/n + 1/N)^{d+1})$ on compact sets. This follows by expanding h_n to order $d-b$ in the term with N^{-b} in (14.14); the finitely many remainders have exactly the asserted total order.

Since $g'(j) \neq 0$, the formal implicit function theorem gives a unique series $v_j(\epsilon, \eta)$ with constant term j satisfying $\mathcal{T}(v_j; \epsilon, \eta) = 0$. Crucially,

$$\mathcal{T}(v; \epsilon, 0) = g(v)R(v; \epsilon).$$

The factor R is a formal unit and g has its simple zero at j . Thus $v_j(\epsilon, 0) = j$, and $v_j - j$ is divisible by η . The formal series

$$B_j(\epsilon, \eta) = \frac{1}{\eta} \left(\frac{1}{v_j(\epsilon, \eta)} - \frac{1}{j} \right) \quad (14.21)$$

therefore contains only nonnegative powers of ϵ, η . All its coefficients are obtained recursively from (14.20). Neither that series nor the implicit series is asserted to converge.

These formal coefficients give genuine asymptotic expansions. Indeed, restrict N/n to a fixed compact subinterval of $(0, \infty)$, which includes $N = n + k - 1$ for fixed k and large n . Truncate the formal root to any prescribed total degree and insert it into the finite expansion of $T_{n,N}$. The residual has the next total order. Locally uniform convergence $T_{n,N} \rightarrow g$, also for its first derivative, and $g'(j) \neq 0$ convert that residual into a root error of the same order. For real parameters this follows directly from the mean value theorem. Computing the root through total order $M + 1$, with residual $O(n^{-M-2})$, gives the height $\log n + N/v$ with error $O(n^{-M-1})$. This accounts explicitly for the factor N introduced by taking the reciprocal.

It remains to determine the polynomial degrees and the first coefficients. Set

$$c = H_{j-1} - \gamma.$$

The local reciprocal-gamma expansion is

$$g(j + \delta) = g'(j)\delta \exp\left(c\delta - \frac{s_2}{2}\delta^2 + \frac{s_3}{3}\delta^3 + O_j(\delta^4)\right), \quad g'(j) = (-1)^j(j-1)!. \quad (14.22)$$

It follows from $g(j + \delta) = (-1)^j\delta \prod_{a=1}^{j-1}(a + \delta)/\Gamma(1 - \delta)$ and the convergent Taylor expansion of $\log \Gamma(1 - \delta)$. In particular,

$$\frac{g''(j)}{g'(j)} = 2c, \quad \frac{g'''(j)}{g'(j)} = 3(c^2 - s_2), \quad \frac{g^{(4)}(j)}{g'(j)} = 4(c^3 - 3cs_2 + 2s_3).$$

Use (14.15) and (14.11) to solve (14.20) through total degree two. The result is

$$\begin{aligned} v_j(\epsilon, \eta) = & j + j^2c\eta + j^2q_1'(j)\epsilon\eta \\ & + (j^3c^2 + j^3s_2 - j^4s_3)\eta^2 + \text{terms of total degree at least three.} \end{aligned} \quad (14.23)$$

For example, the η coefficient is $-\mathcal{D}_1g(j)/g'(j) = j^2c$. At $\epsilon = 0$, the coefficient of η^2 is

$$-\frac{\frac{1}{2}g''(j)(j^2c)^2 + (\mathcal{D}_1g)'(j)j^2c + \mathcal{D}_2g(j)}{g'(j)},$$

which simplifies to the displayed value. At first order in η , replacing g by gR replaces c by $c + \partial_v \log R(j; \epsilon)$, giving the $\epsilon\eta$ term. Consequently

$$B_j(\epsilon, \eta) = -c - (j + \frac{1}{2})\epsilon + (j^2s_3 - js_2)\eta + \text{terms of total degree at least two.} \quad (14.24)$$

Now put $\kappa = k - 1$ and substitute

$$\eta = \frac{\epsilon}{1 + \kappa\epsilon}. \quad (14.25)$$

This defines the polynomials $C_{j,m}$ by

$$B_j\left(\epsilon, \frac{\epsilon}{1 + \kappa\epsilon}\right) = -c + \sum_{m \geq 1} C_{j,m}(\kappa)\epsilon^m. \quad (14.26)$$

Every nonconstant monomial $\epsilon^a\eta^b$ has $a + b \geq 1$. If $b \geq 1$, its contribution to the coefficient of ϵ^m is a constant times κ^{m-a-b} , and if $b = 0$ it is independent of κ . Therefore $\deg C_{j,m} \leq m - 1$. For $m \geq 2$, the only possible term of degree $m - 1$ comes from η itself; by (14.24) its coefficient is (14.18). Since $s_3 = -\sum_{a=j}^{\infty} a^{-3} < 0$ and $s_2 > 0$, this coefficient is nonzero. At $m = 1$ the constant polynomial is $C_j = j^2s_3 - js_2 - j - 1/2 < 0$, also nonzero. This proves the exact degree assertions and the stated formulas. The formal-to-asymptotic argument above completes (14.16). $\square$

Corollary 14.4 (The first two largest-root corrections). *For every fixed positive integer k ,*

$$X_{n,k,1} = n + k - 1 + \log n + \gamma + \frac{C_0(k)}{n} + \frac{C_1(k)}{n^2} + O_k(n^{-3}), \quad (14.27)$$

where, in Cohen's indexing,

$$\begin{aligned} C_0(k) &= -\frac{3}{2} - \zeta(2) - \zeta(3), \\ C_1(k) &= (k-1)(\zeta(2) + \zeta(3)) + \zeta(2)^2 - \zeta(2)\zeta(3) + 2\zeta(3) + 3\zeta(5) - \frac{1}{12}. \end{aligned} \quad (14.28)$$

Thus $C_0(k) = -4.346990970007820721872\dots$ and $C_1(1) = 6.160067472396514377053\dots$

Proof. The first identity is (14.17) at $j = 1$. For the second, the gamma-ratio coefficient following q_1 is $q_2(v) = v(v+1)(v+2)(3v+1)/24$. Write, near $v = 1$,

$$\frac{g(1+z)}{g'(1)} = z \exp \left(cz - \frac{s_2}{2}z^2 + \frac{s_3}{3}z^3 - \frac{s_4}{4}z^4 + \frac{s_5}{5}z^5 + O(z^6) \right),$$

where $c = -\gamma$, $s_2 = \zeta(2)$, $s_3 = -\zeta(3)$, $s_4 = \zeta(4)$ and $s_5 = -\zeta(5)$. Substitute q_1, q_2 and $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$ into (14.20), then use (14.25). Equating three successive coefficients in the simple-root equation and taking the reciprocal in (14.21) gives

$$C_1(k) = -s_2^2 + s_2s_3 + (k-1)(s_2 - s_3) - 2s_3 + 5s_4 - 3s_5 - \frac{1}{12}.$$

The formal symbol c cancels. Since $5\zeta(4) = 2\zeta(2)^2$, this is (14.28). This is a finite coefficient calculation by the recursion already specified in (14.20); a rational-arithmetic replay is included among the accompanying verification artifacts. The remainder follows from Theorem 14.3 with $M = 2$. $\square$

Corollary 14.5 (Arithmetic form of every correction coefficient). *For fixed positive integers j, m ,*

$$C_{j,m}(\kappa) \in \mathbb{Q}[\zeta(2), \zeta(3), \dots, \zeta(2m+1)][\kappa].$$

Thus Euler's constant is confined to the leading shift in (14.16); it is absent from every subsequent coefficient produced by the construction.

Proof. Fix j and set $c = H_{j-1} - \gamma$. Extract one more exponential, writing $h_n(v) = e^{cv}a_n(v)$. Exactly as in (14.7),

$$T_{n,N}(v) = (1 + cv/N)^N \frac{N!}{N^N} [z^N] e^{Nz} a_n(sz), \quad s = \frac{v}{1 + cv/N}.$$

The height is now $X = \log n - c + N/s$. Up to the irrelevant nonzero constant $e^{-cj}g'(j)$, the local asymptotic amplitude at j is

$$\delta \exp \left(\sum_{r \geq 2} \frac{(-1)^{r-1} H_{j-1}^{(r)} - \zeta(r)}{r} \delta^r \right) R(j + \delta; \epsilon).$$

The finite harmonic sums are rational and the coefficients of R are rational polynomials. The recursion for a height coefficient of order n^{-m} uses the root through total degree $m+1$, hence amplitude derivatives of order at most $2m+2$. The initial zero factor δ reduces the largest needed zeta index to $2m+1$. After the displayed normalization, all divisions in the simple-root recursion are by nonzero rational numbers. This proves the claim. $\square$

The theorem proves the fixed- j assertion in Cohen's Conjecture 7.1(2), since $H_n = \log n + \gamma + O(1/n)$, and proves his full all-orders assertion in Conjecture 7.1(4) for every fixed positive integer k . In his indexing the coefficient called $C_m(k)$ is our $C_{1,m+1}(k-1)$, which has degree exactly m . The first numerical constant proposed there is identified exactly in Corollary 14.4. The largest-root estimate, together with real-rootedness, also gives Conjecture 7.1(1), and Proposition 14.1 gives Conjecture 7.1(3). These are statements about the precise conjectures in the cited version; no exhaustive priority claim is intended. The derivative interpretation proposed for $k \leq 0$ is not needed here. The theorem is a fixed- j , fixed- k result: its remainder constants are not asserted uniform for growing indices.

The bulk law and the edge theorem describe different scales. A positive fraction of the zeros stays in a bounded interval of the logarithmic coordinate $X = L$, whereas every fixed number of largest zeros grows linearly with n . The L^{-1} tail of the bulk limiting measure agrees with the leading edge scale n/j . A uniform transition for $j \rightarrow \infty$ with $j/n \rightarrow 0$, and a regime in which k also grows, remain natural research problems.

15 A Bessel limit for the smallest roots

The fixed largest roots have size n/j , while the bulk law leaves the smallest roots unresolved. The opposite endpoint has a different, explicit scaling limit. It follows directly from the same Stirling coefficient formula, and completes the leading description of the three root regimes considered here.

For $n, k \geq 1$, use the harmless scalar normalization

$$P_{n,k}(X) = \frac{(-1)^{n+k+1} L_{n,k}(X)}{k! \binom{n+k-1}{n}} = \sum_{j=1}^n (-1)^j \begin{bmatrix} n \\ n-j+1 \end{bmatrix} \frac{X^{k-1+j}}{(k-1+j)!}. \quad (15.1)$$

In particular $P_{n,1} = p_n$. Let $z_{k,j}$ be the j th positive zero of the Bessel function J_k ; these zeros are simple [15].

Theorem 15.1 (An entire Bessel scaling limit). *For each fixed positive integer k , locally uniformly for complex x ,*

$$n^{2k} P_{n,k}(x/n^2) \longrightarrow -(2x)^{k/2} J_k(\sqrt{2x}). \quad (15.2)$$

The right side is understood by its entire power series, so it has no square-root branch ambiguity. If $x_{n,k,j}$ denotes the j th smallest positive root of $L_{n,k}$, then for every fixed $j, k \geq 1$,

$$n^2 x_{n,k,j} \longrightarrow \frac{z_{k,j}^2}{2}. \quad (15.3)$$

For the diagonal zero-velocity lattice parameters of section 13, this says

$$n^2 \log A_{n,j} \longrightarrow \frac{z_{1,j}^2}{2}, \quad n^2 (A_{n,j} - 1) \longrightarrow \frac{z_{1,j}^2}{2}. \quad (15.4)$$

Proof. Write $m = j - 1$. The standard product for unsigned Stirling numbers gives

$$\begin{bmatrix} n \\ n-m \end{bmatrix} = e_m(1, 2, \dots, n-1),$$

where e_m is the elementary symmetric function of degree m . For each fixed m ,

$$n^{-2m} \begin{bmatrix} n \\ n-m \end{bmatrix} \longrightarrow \frac{1}{2^m m!}. \quad (15.5)$$

Indeed $(\sum_{r=1}^{n-1} r)^m$ is $m!e_m$ plus terms with repeated indices. For $m \geq 2$ the sum of the latter is bounded by

$$\binom{m}{2} \left(\sum_{r=1}^{n-1} r^2 \right) \left(\sum_{r=1}^{n-1} r \right)^{m-2} = O_m(n^{2m-1}).$$

The cases $m = 0, 1$ are direct, and $n^{-2} \sum_{r=1}^{n-1} r \rightarrow 1/2$. For all m, n the same positive expansion gives the useful bound

$$0 \leq n^{-2m} \left[\begin{matrix} n \\ n-m \end{matrix} \right] \leq \frac{1}{2^m m!}, \quad (15.6)$$

with the coefficient interpreted as zero for $m \geq n$.

Substituting (15.1) yields

$$n^{2k} P_{n,k}(x/n^2) = - \sum_{m=0}^{n-1} (-1)^m n^{-2m} \left[\begin{matrix} n \\ n-m \end{matrix} \right] \frac{x^{k+m}}{(k+m)!}.$$

On every compact set, (15.6) gives an absolutely summable common bound for its terms. Termwise passage to the limit proves

$$- \sum_{m \geq 0} \frac{(-1)^m x^{k+m}}{2^m m! (k+m)!} = -(2x)^{k/2} J_k(\sqrt{2x}),$$

the defining Bessel series [15], and proves local uniformity.

The limiting function has a zero of multiplicity k at zero and simple positive zeros $z_{k,j}^2/2$. Apply Rouché's theorem on disjoint small circles around zero and the first finitely many positive zeros. The polynomial on the left has exactly multiplicity k at zero, and its remaining zeros are real, simple, and positive by the operator theorem in section 14. Uniform nonvanishing on the compact real intervals between those circles therefore identifies the first positive polynomial roots with the respective limiting zeros; no additional small root can enter at zero. This proves (15.3). For $k = 1$, $x_{n,1,j} = \log A_{n,j}$. Since this tends to zero, $e^x - 1 \sim x$ gives (15.4). $\square$

For example, $z_{1,1}^2/2 = 7.340985321 \dots$. This endpoint constant is not obtained by inserting a fixed j into the bulk quantile formula: the bulk theorem assumes $j/(n-1)$ stays away from both endpoints. A uniform bridge between the Bessel regime, the angle-distributed bulk, and the largest-root expansion remains a substantive further problem.

16 Exact certificates, numerical diagnostics, and reproducibility

The accompanying computations have two distinct purposes. Exact rational and finite-field calculations verify finite identities and isolate specified algebraic numbers. High-precision quadrature and coefficient evaluation illustrate analytic asymptotics whose proofs are given in the preceding sections. The latter calculations do not certify continuation of an inverse sheet or optimization over an unbounded parameter domain.

16.1 Finite algebra that can be replayed exactly

The verifier `code/verify_kernel_algebra.py` uses only Python's standard library and rational arithmetic. It reconstructs the two derivative numerators of K_2 , computes their Sylvester resultant, checks the elimination substitutions, obtains the primitive polynomial for M_2 , and proves its

irreducibility modulo five. It also checks the rational interior sample and isolates the relevant roots. Its outward rational interval calculations certify

$$1.1189386859610526747774667909001084300609 < M_2, \\ M_2 < 1.1189386859610526747774667909001084300610.$$

The full rational endpoints, rather than only their decimal displays, are retained in `data/kernel_algebra.json`.

The independent verifier `code/verify_cohen_c1.py` also uses only standard-library rational polynomial arithmetic. It verifies the exact coefficient-extraction moments through the order needed for the first two largest-root corrections, performs the perturbation algebra, and checks cancellation of the Euler-constant variable. The zeta-value formula is obtained by exact substitution and $\zeta(4) = \frac{2}{5}\zeta(2)^2$, not by an integer-relation fit. The record is `data/cohen_c1_certificate.json`.

Two further checks compare independently generated polynomial formulas. `code/verify_diagonal_inverse.py` compares the exact Stirling and nonlinear inverse-series recurrences through degree 35, and uses rational enclosures of $\log 2$ to certify the corresponding finite signs. `code/verify_cohen_edges.py` checks the generalized operator normalization for $1 \leq n \leq 14$, $1 \leq k \leq 5$ and symbolically verifies the general first coefficient $C_{j,1}$. These finite checks are useful independent replays; the all-index identities and asymptotic results have separate proofs.

16.2 Large-index axis diagnostics

Direct gamma quadrature evaluates $R_N(0, b) - 1$ without subtracting large finite Euler sums. The implementation forms

$$\exp\left(\log(1+v) + (N-1)\log(1-v^2) - \log(1+v^2)\right) - 1$$

with stable logarithm and exponential differences, then integrates against the gamma density. The maximizing order is found by the gamma-score derivative; negative curvature is checked numerically. The sample below uses 65 decimal digits of working precision.

N	Axis maximizing b_N	$(C_N^{\text{ax}} - 1)/(cN^{-p})$
10^2	12.3318927895	1.0759792475
10^4	23.7920662690	1.0296132033
10^6	35.1917432654	1.0086616486
10^8	46.5618035683	1.0025176685

Table 1: Numerical axis diagnostics. The finite numbers are not used to choose an unproved onset threshold for global axis reduction.

The second Euler scale deserves particular care: the proved coefficient $T(B_0) \approx -2098.43$ makes the approach to its leading amplitude slow. The data at $N \leq 10^8$ are consistent with the first asymptotic law; they are not a useful basis for fitting the much more delicate next coefficient. The asymptotic proof identifies that coefficient through a Mellin integral and uniform domination.

To resolve this slow approach numerically, the additional script `code/transition_remainder_diagnostics.py` evaluates the positive Taylor remainder $\varepsilon_N(\hat{b}_N)$ after dividing by the gamma density at $t = \frac{1}{2} \log N$. It forms no array of length N and uses stable exponential and hypergeometric remainders. Define the dimensionless diagnostic

$$Q_N = \frac{\varepsilon_N(\hat{b}_N)}{D_0 N^{-\mu} / \sqrt{\log N}}.$$

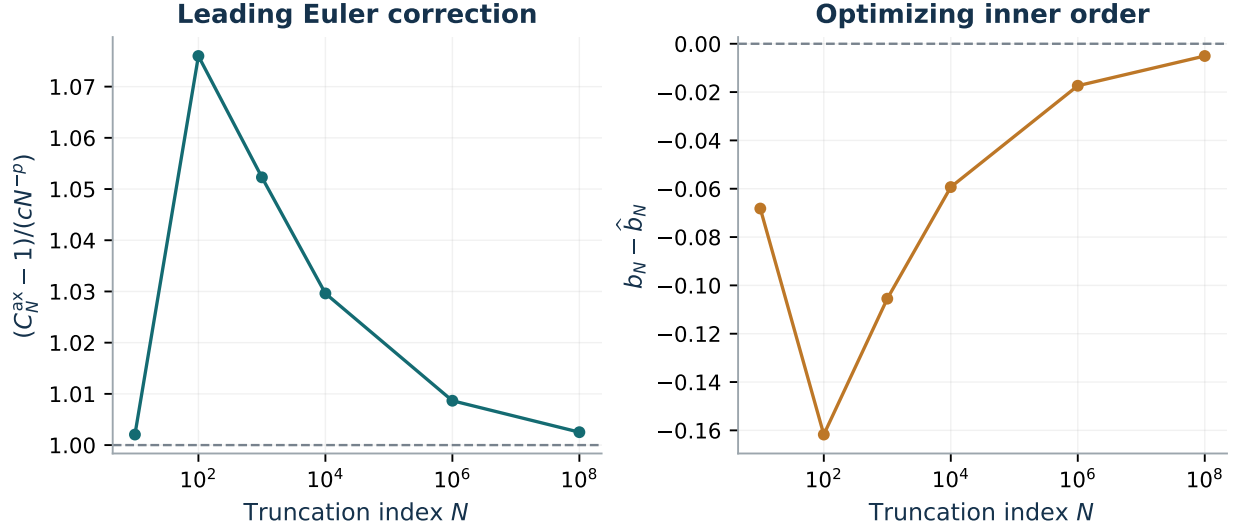


Figure 1: The optimized axis remainder approaches the proved leading law, while the inner-order displacement tends to zero. The coefficient in its finer inverse-logarithmic expansion is large, so moderate indices need not display the final second-scale amplitude accurately.

The theorem gives $Q_N \rightarrow 1$ and $(\log N)(Q_N - 1) \rightarrow T(B_0)$. The following 75-digit quadratures show how late these limits become numerically visible.

$\log_{10} N$	Q_N	$(\log N)(Q_N - 1)$
8	0.1389757434	-15.860653
100	0.3477552643	-150.184901
1000	0.6776397628	-742.261877
10000	0.9271461038	-1677.522954
100000	0.9911352486	-2041.184436
1000000	0.9990912540	-2092.464968
Proved limit	1	-2098.428735

Table 2: Transition-remainder diagnostics at the leading center $\hat{b}_N$. These evaluate the positive remainder at the stated center. Large exponents are handled through logarithms and normalized quadrature.

At $N = 10^8$, an independent direct-axis calculation minus the exact first Taylor terms agrees with this normalized remainder to relative error below 10^{-34} . This is a numerical cross-check of the implementation, not an interval certificate for the displayed quadratures. The slow approach reinforces the need for the analytic proof of the second scale.

16.3 Pointwise kernel and inverse-series checks

The kernel calculations solve the two stationary equations in the scaled coordinates $(c, y) = (Nr, y)$ with 85-digit arithmetic. At $N = 2$ the exact certificate independently proves global optimality. For other particular finite indices the table records stationary candidates; the article’s global uniqueness theorem is eventual.

The inverse-series diagnostics use 360-digit arithmetic for $A = 1.1, 2, 4, 10$ through index 240.

N	Nr	y	Stationary kernel value
2	1.7045169541	0.2527026784	1.1189386860
4	2.3345294376	0.2084961852	1.0878146577
10	3.0411292987	0.1758872724	1.0689106048
100	3.8193399906	0.1489190557	1.0555582496
10000	3.9397958507	0.1453408168	1.0538792067

Table 3: Stationary-kernel diagnostics from the explicit rational kernel, distinct from the order-averaged constants.

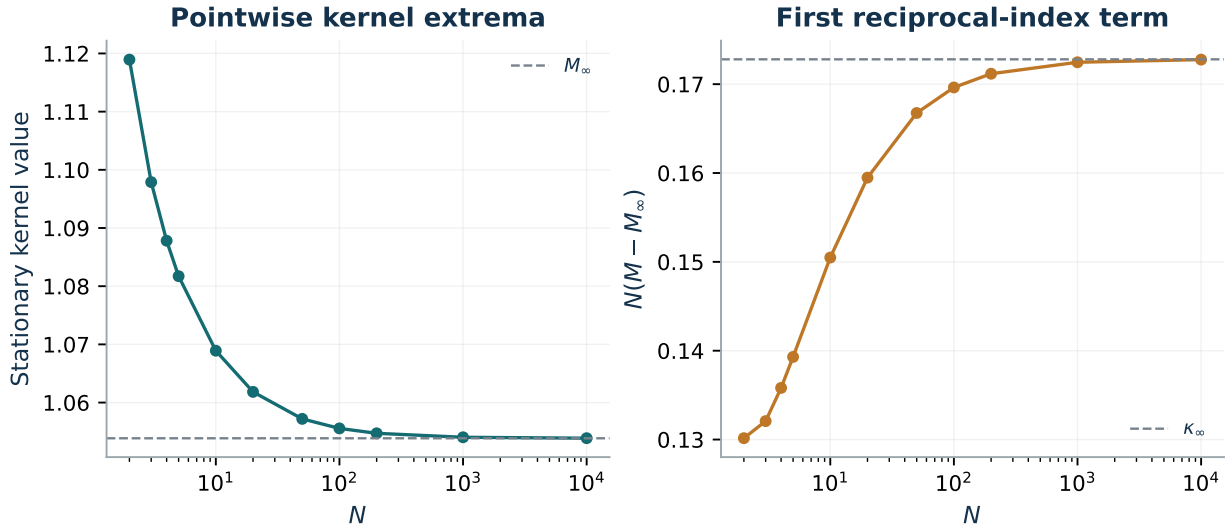


Figure 2: Numerical stationary values and their first scaled excess. The dashed levels are the analytically characterized M_∞ and κ_∞ . The proof of convergence covers all global maximizers and does not depend on this stationary-point computation.

Direct Stirling polynomial evaluation and the independent coefficient recurrence agree to more than 150 relative decimal digits in every tested case. The record also compares the exact barrier factorization at separate points, the first two coefficient models, the boundary sums, and the renormalized derivative sum. An error estimate involving a potentially vanishing cosine is always displayed in additive normalized form.

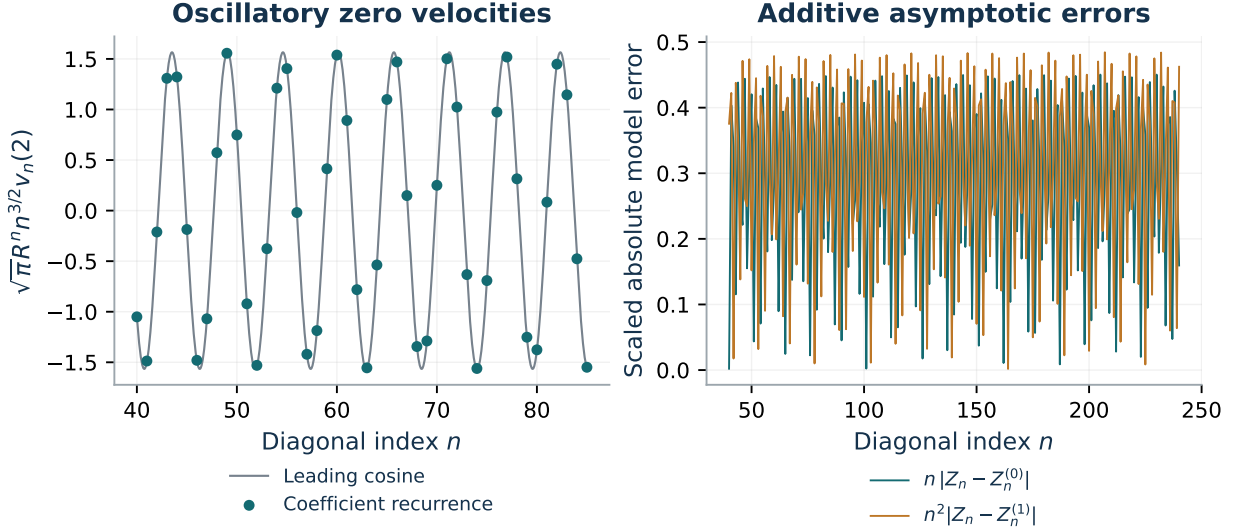


Figure 3: At $A = 2$, coefficients from an independent recurrence follow the proved oscillatory phase. The right panel plots the errors after the scalings predicted by the leading and corrected additive asymptotics. These are finite numerical diagnostics of formulas whose sheet selection and transfer hypotheses are proved analytically.

16.4 Lambert roots at the bulk and at both endpoints

For degrees 12, 30, and 60, the script `code/compute_bulk_root_examples.py` isolates every real root of the exact rational polynomial by symbolic rational interval arithmetic. Each positive-root interval has width at most 10^{-14} ; all intervals have multiplicity one. Their midpoints are used only for plotting. The distribution function is compared with the exact angle law, and scaled polynomials are compared with the Bessel limit.

At the opposite endpoint, high-precision evaluation of the reciprocal polynomial avoids its very large leading powers. The root diagnostics cover 45 combinations of fixed j, k and $64 \leq n \leq 1024$. For $j = k = 1$, define

$$B_n = n(X_{n,1,1} - n - \log n - \gamma), \quad D_n = n^2(X_{n,1,1} - n - \log n - \gamma - C_0/n),$$

where $C_0 = -\frac{3}{2} - \zeta(2) - \zeta(3)$. The next table compares these quantities with their proved limits.

The file `data/figure_samples.json` also records finite tests of the newly proposed indexed bulk-phase formula in [conjecture 18.3](#). That conjecture is deliberately kept out of every theorem dependency.

16.5 Rebuilding and rechecking the package

The main source is supplied both as `article.tex` and as modular sections with `main.tex`. Run `python3code/assemble_article.py` after editing modules, then `latexmk-pdf-halt-on-error`

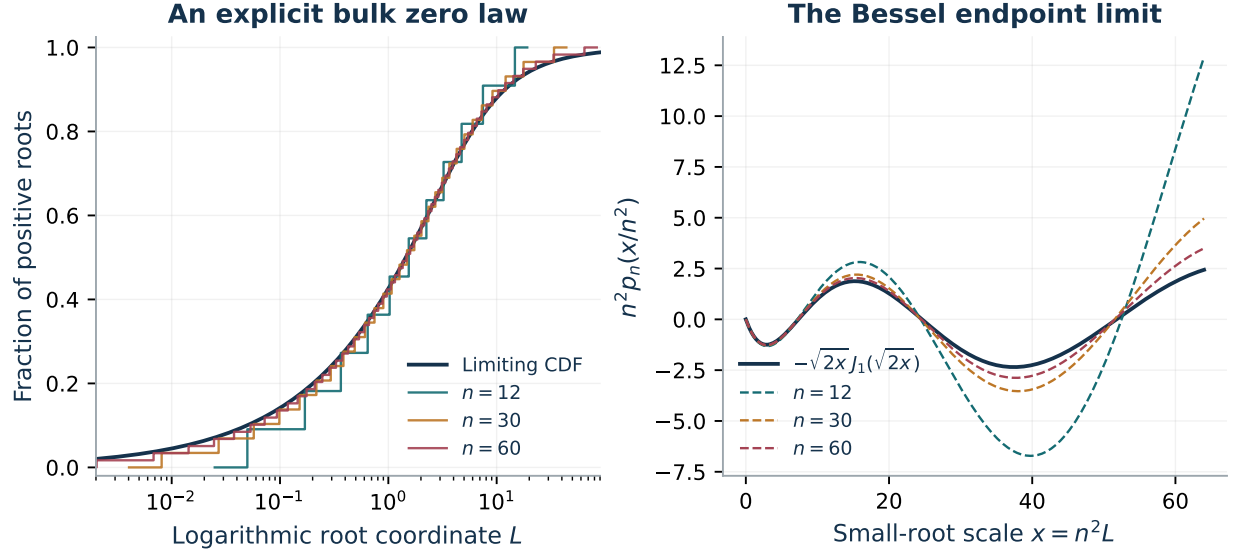


Figure 4: Left: finite positive-root distribution functions, based on exact rational isolation, approach $\theta(L)/\pi$. Right: the polynomial scaling at the small-root endpoint approaches the entire Bessel function in theorem 15.1. The logarithmic axis on the left makes both endpoint regions visible.

n	B_n	D_n
64	-4.25307307280	6.01074542118
128	-4.29945689430	6.08436169117
256	-4.32307710966	6.12194824995
512	-4.33499694535	6.14094062226
1024	-4.34098463490	6.15048715477
Proved limit	-4.34699097001	6.16006747240

Table 4: Largest-root diagnostics. The limit constants come from exact symbolic formulas and the all-orders theorem.

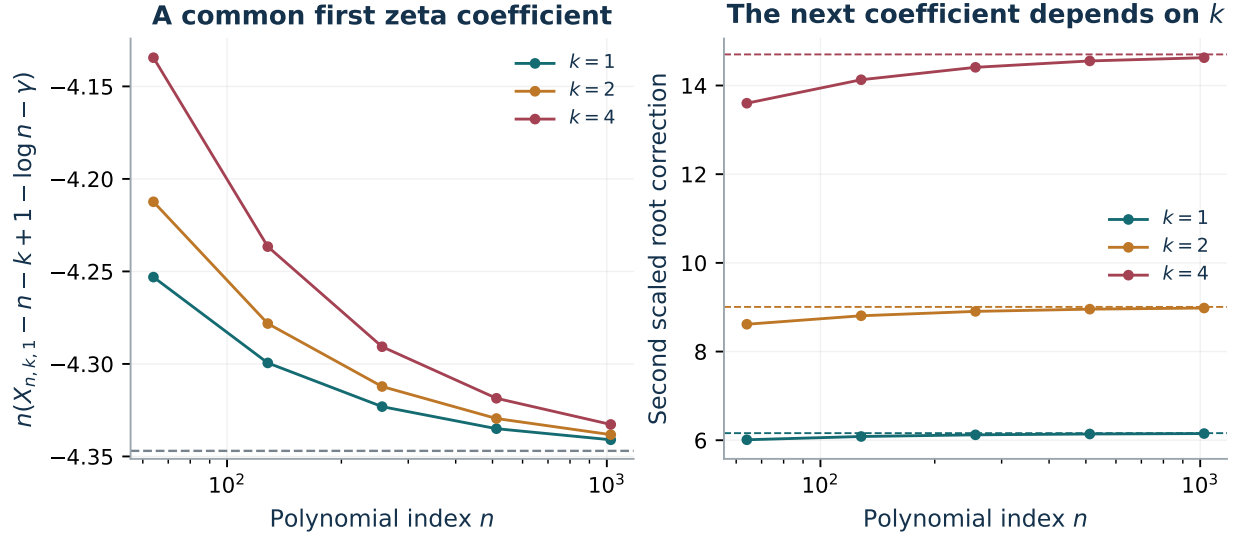


Figure 5: The first largest-root correction is independent of k . The next one is affine in $k - 1$, with the exact expression proved in section 14; its respective limiting levels are dashed. The numerical root residuals are separate from the analytic root ordering and remainder proof.

`ticle.tex`. The file `code/reproduce.py` replays the exact checks by default; optional flags replay the more expensive diagnostics and figures. A plain-text readme lists the individual commands and the versions used. All scripts operate on local files and require no repository write access or network connection.

The provenance directory pins the source revision and gives Git blob hashes for retrieved material. A package checksum manifest records the delivered files. The PDF was compiled from the supplied consolidated TeX and visually checked after rendering, with particular attention to long formulas, tables, cross-references, and the distinction between the proved limits and finite diagnostics.

17 Audit findings and proposed manuscript amendments

The source review found several statements whose status can now be strengthened, two important antecedents that should be cited, and a small notation error. It did not find a counterexample to the canonical theorems used as inputs here. In particular, the source material itself correctly warns that formal critical values do not select an inverse sheet, that local kernel convergence does not prove convergence of global maxima, and that floating-point agreement is not an arithmetic identity proof. Those distinctions should be preserved when the present results are integrated.

17.1 Changes in theorem and conjecture status

Source target	Proposed status and supporting result
Uniform-bounds report, <code>conj:euler-rate</code>	Promote the matching global asymptotic to a theorem. Cite theorems 3.1, 3.9 and 4.1 for eventual exact axis reduction, the complete near-optimality criterion, and the sharper next term.
Sharp-Euler report, <code>conj:limit</code>	Promote to a theorem with the quantifier covering every sequence of global maximizers. The proof is theorem 6.3; theorem 6.4 adds analytic perturbation and eventual uniqueness and decrease.
Sharp-Euler report, <code>conj:maxima</code>	Retain the all-index conjecture. Its uniqueness and interiority assertion is proved for $N = 2$; uniqueness and strict decrease are proved for all sufficiently large N . No verified finite threshold covers every remaining index.
The proposed K_2 elimination target	Mark completed. Theorem 7.1 proves a unique global interior maximum and its irreducible degree-seven equation, with an exact rational certificate.
Harmonic continuation, <code>diag:q:dominance</code>	Promote from a conjecture at $A = 2$ to the selected-sheet theorem for every $A > 1$, theorem 9.1. Include the square-root sign and the radial-slit continuation domain.
Harmonic continuation, <code>diag:q:</code> <code>conditional-asymptotic</code>	Remove the conditional hypothesis after the selected-sheet theorem has been inserted. Credit the existing local calculation and retain the additive form of the error.
Diagonal coefficient polynomials	Identify them explicitly as Cohen's Lambert polynomials up to a sign. Add the bulk law, largest-root expansion, and Bessel endpoint limit as further results; do not present the coefficient array itself as newly discovered.

The independent maximum on the axis is unique and nondegenerate at every integer index, by proposition 3.6, and its continuous interpolation is strictly log-convex. This does not settle all-index axis reduction for the two-parameter global problem. Similarly, the analytic dependence of the *kernel* optimum on $1/N$ is a convergent expansion, whereas the all-orders Lambert-root expansion is an asymptotic expansion. Their convergence statuses must remain distinct.

17.2 Two substantive attribution corrections

The Lambert inverse series. In the discussion following the exact generating function, add the map $\lambda = \log A$, $\sigma = -t/A$, $v = V_A(t)$ to Kalugin and Jeffrey's equation $1 - e^{-v} + \sigma v - \sigma \lambda = 0$ [5].

Their fixed- λ series and radius criterion are antecedents of the coefficient and radius formulas here. The repository’s missing selected-sheet proof is supplied by the barrier argument; it should not be described as the first occurrence of the radius formula.

The Lambert coefficient polynomials. Add $L_n(X) = (-1)^n p_n(X)$ and the generalized normalization from Cohen’s Proposition 4.8 [6]. The real-rooted operator argument already present in the harmonic report is directly relevant to Cohen’s conjecture. The present interlacing and extreme-root proofs make the comparison explicit. When quoting the numerical constant in Cohen’s original question [7], distinguish the two normalizations:

$$\begin{aligned} X_{n,1,1} &= n + \log n + \gamma - \frac{\frac{3}{2} + \zeta(2) + \zeta(3)}{n} + O(n^{-2}), \\ &= n + H_n - \frac{2 + \zeta(2) + \zeta(3)}{n} + O(n^{-2}). \end{aligned}$$

The difference of $1/2$ in the coefficients comes from $H_n = \log n + \gamma + 1/(2n) + O(n^{-2})$; it is not a discrepancy in the root asymptotic.

17.3 A verified notation cleanup

In the proof of `rw:thm:transport` in `chapters/05-real-positive-kernel.tex`, the instruction to use the Hurwitz formula “with $q = n$ ” should read “with $x = n$ ”. Here x is the Hurwitz argument; q is the regularizing kernel used in the neighboring formula. The following sentence refers to boundedness of h , but the regularizer was named $q(t)$, so that occurrence should read boundedness of q . These are notation corrections and do not change the transport identity or its proof. The package includes a small unified diff against the pinned source for these two replacements.

17.4 Integration boundaries and verification status

The integration notes identify exact source paths and labels, suggest a placement for the new sections, and list the accompanying verifier commands. They do not modify the remote repository. The assembled article is also supplied as modular TeX to make relocation of sections and theorem environments straightforward.

The exact checks prove finite polynomial identities, finite-field irreducibility, rational interval enclosures, and the displayed finite coefficient cancellations. They do not formally verify Rouché’s theorem, the probability representation, singularity transfer, or asymptotic uniformity. Those claims have the complete mathematical proofs in the article and the separate review notes in the package. The source scan and targeted primary-literature comparison support the stated continuation of this repository; they are not an exhaustive bibliographic priority search.

18 Further research questions and explicit conjectures

The results isolate several precise next problems. The first group would complete finite-parameter statements already suggested by the source reports. The later groups ask for uniform transitions or new structures beyond the results proved here.

18.1 Complete the finite-index Euler extremum problem

Conjecture 18.1 (Axis reduction at every truncation). *For every integer $N \geq 1$,*

$$C_N = \max_{b>0} R_N(0, b),$$

and the unique maximizer on $a \geq 0, b > 0$ lies on that axis.

The case $N = 1$ is part of the known universal-constant theorem; the present article proves all sufficiently large N . The conjecture is about a global maximum over both orders. It should not be replaced by a pointwise comparison $R_N(a, b) \leq R_N(0, b)$ at every fixed b : that stronger statement is not the claim and has no such proof here. A productive approach would combine exact small-index analysis with an effective, moderately sized threshold in the localization argument. The relevant difficulties occur outside the eventual competitive band.

An explicit threshold N_0 is itself useful. The present split at $b = 2 \log N$ produces a slowly vanishing factor $N^{1-2 \log(5/3)}$. Improving this estimate, perhaps with a more carefully optimized gamma lower-tail bound, would turn the eventual theorem into a practical enclosure theorem. A finite numerical scan without a proved tail threshold would not close this question.

Conjecture 18.2 (All-index kernel uniqueness and decrease). *For every $N \geq 2$, K_N has a unique global maximizer in $(0, 1)^2$ and $M_{N+1} < M_N$.*

This restates the remaining portion of the sharp-Euler report's conjecture. The exact $N = 2$ elimination and the analytic branch near $1/N = 0$ suggest combining algebraic continuation with certified boundary exclusions. Monotonicity of the envelope B_N alone does not prove monotonicity of M_N .

18.2 Refine the gamma-transition asymptotics

The next Euler scale has the form $N^{-\mu}(\log N)^{-1/2}$ and carries a large first inverse-logarithmic coefficient. Determine a uniformly useful approximation across the moderate- N regime and the true asymptotic regime. The exact positive remainder integral is a better starting point than subtraction of nearly equal Euler sums. One target is a resummed transition integral whose error remains explicit while retaining the slowly decaying tail responsible for the large coefficient.

For other residual kernels, moment competition may put the analogous Mellin exponent at an integer. Classify when logarithmic resonances replace a pure fractional-power correction, and determine the resulting maximizer displacement. The present proof suggests a general method: identify the competing exponential moments, localize the parameters, and analyze the positive remainder near the moving kernel transition. The exponent and amplitude need to be derived anew for each kernel.

The joint profile in theorem 3.9 gives the exact first loss from leaving the axis. Higher joint terms could quantify stability of near-optimal order selection at prescribed precision. They should be stated on explicit joint scales for $a, b - \hat{b}_N$, and N ; pointwise small- a and large- N expansions are not automatically uniform.

18.3 Determine the arithmetic of the oscillation phase

For a fixed algebraic value such as $A = 2$, is $\theta(A)/(2\pi)$ irrational? The article proves sign-density statements in both the rational and irrational cases, so no unproved phase arithmetic is needed for those theorems. A proof of irrationality at $A = 2$ would give density $1/2$ for each sign and would make rotation methods available for more precise gap statistics.

Find the least eventual block length that must contain both signs. The bound $\lfloor \pi/\theta \rfloor + 2$ is explicit and gives four at $A = 2$. Whether a smaller block can work is a question about the actual rotation orbit and its distances from the sign boundaries, together with the coefficient error. Likewise, quantify the number of indices where the normalized velocity is unusually small. A relative cosine approximation is unsuitable for these exceptional indices; the all-order additive expansion is the available tool.

The finite-part identity for the critically weighted first derivative can be extended to higher derivative sums. Determine their full subtraction polynomials in fractional powers of the cutoff, and identify which finite constants are local Puiseux coefficients. This should give explicit renormalized Stirling identities, with ordinary convergence and regularized limits carefully distinguished.

18.4 A proposed bulk quantization formula

Let $L_{n,j}$ be the positive zeros of p_n in increasing order, $\theta_{n,j} = \theta(e^{L_{n,j}})$, and evaluate u_* , χ , χ_1 at $A = e^{L_{n,j}}$. Put $\phi = -\arg \chi$ as before. The following indexed refinement is consistent with the finite root isolations supplied in the package.

Conjecture 18.3 (An indexed phase formula in the bulk). *For every fixed $\epsilon \in (0, 1/2)$, uniformly for $\epsilon(n-1) \leq j \leq (1-\epsilon)(n-1)$,*

$$n\theta_{n,j} + \phi(\theta_{n,j}) = \pi \left(j + \frac{1}{2} \right) + \frac{1}{n} \Im \left(\frac{\chi_1}{\chi} \right) + O_\epsilon(n^{-2}). \quad (18.1)$$

The unproved content includes the global root indexing, in particular the exact shift $j + 1/2$. Local transfer locates roots near a half-integer phase and gives the displayed correction once the index is identified; the bulk counting theorem alone does not fix that index. One possible route is to match the Bessel endpoint asymptotic to a uniform interior expansion in an overlap where $j \rightarrow \infty$ and $j/n \rightarrow 0$. This is a conjectural asymptotic identity, not a formula used anywhere in the proofs of this article.

18.5 Connect the smallest, bulk, and largest root regimes

The three proved regimes are

$$L_{n,j} \asymp n^{-2} \quad (j \text{ fixed from the small end}), \quad L_{n,j} = O(1) \quad (j/n \text{ in the bulk}),$$

and $X_{n,k,j} \sim n/j$ for j fixed from the large end. Develop uniform expansions when the root index grows while remaining $o(n)$. At the large end, the main question is how the expansion around the j th zero of $1/\Gamma(1-w)$ changes when j is no longer fixed. At the small end, the Bessel zero expansion should be matched to the singularity-transfer phase. Either transition would sharpen the weak zero law and give useful discrepancy estimates.

The all-orders largest-root theorem treats fixed positive integer k . Determine uniform ranges of $k = k(n)$, and carefully extend the theorem to Cohen's derivative interpretation for nonpositive k if desired. The exact logarithmic change of variables remains a promising starting point, but the compact-parameter gamma expansion used here does not justify growing- k or growing- j conclusions.

Recent polynomial-profile methods [16] may supply another description of the bulk law. Determine whether the Lambert polynomial family admits a useful limiting Cauchy transform, free-convolution interpretation, or a potential-theoretic variational principle. Any such identification should explicitly handle its degree-dependent differential operator; the fixed-function examples in the general theory are not automatic substitutes.

18.6 Return the analytic tools to polylogarithmic identities

The exact error kernels remain available as certified numerical tools for cyclotomic and Gaussian polylogarithm identities. The unresolved mixed S_6 reduction is an appropriate arithmetic target, but a candidate relation requires an independent reduction mechanism such as iterated-integral relations, a differential identity with controlled boundary values, or a valid coaction argument. A high-precision integer relation search alone would still leave a conjecture.

For roots of unity beyond i , seek a sector or matrix positivity statement that replaces the scalar order of the present Gaussian kernel. At higher depth, identify the correct iterated compensated differences and prove integrability at each step. The all-depth oscillation results already in the incoming material concern a different part of this problem; they should be used with their stated hypotheses, not treated as a ready-made extension of the present two-order Euler extremum theorem.

Finally, the exact finite components provide natural formalization targets: the Euler coefficient identity, the rational derivative and resultant certificates, finite-field irreducibility, the elementary real-rooted operators, and the algebra of the first edge coefficients. A later formal development could then connect them to gamma integration, Rouché continuation, and singularity transfer. The package supplies reproducible ordinary proofs and finite arithmetic; that proposed formal bridge remains future work.

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